

THINK Ahead: An eight-chart Easter Eggstravaganza

Fun for all the family this week as James Smith goes on an Easter hunt for some cracking charts to eggs-plain the current economic crisis



The economic impact of energy prices remaining elevated for longer isn't quite the Easter surprise we were hoping for

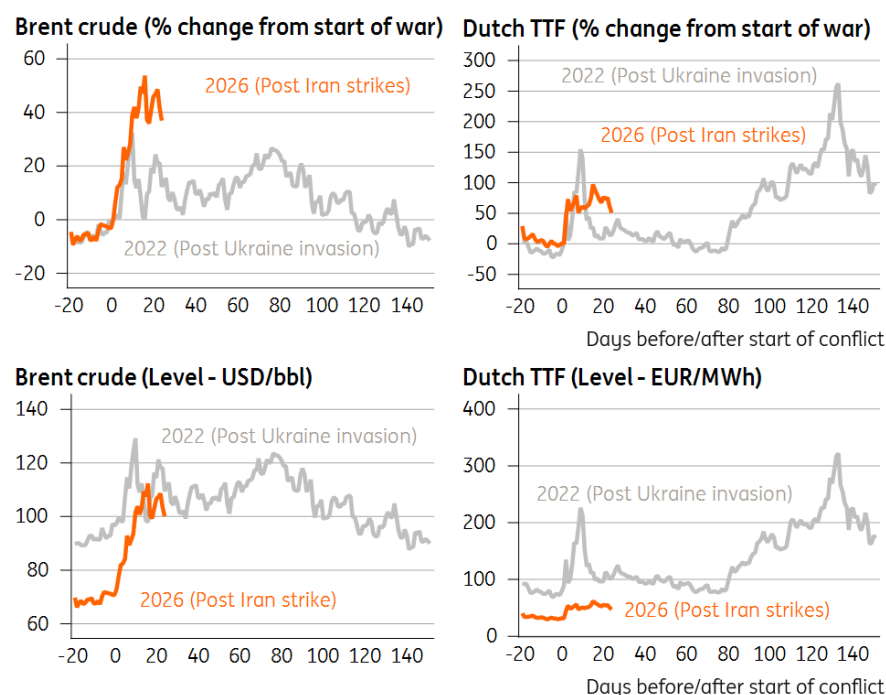
1 This energy crisis is better and worse than 2022

One month into the Iran war and it's worth taking stock.

The good news, to the extent that there is any, is that the level of the price shock isn't as bad as it was at the same point in the Ukraine conflict. Oil prices are still below the 2022 peak of 123 USD/bbl - about 140 in today's money, adjusted for inflation. And helpfully for Europe, gas prices remain much lower. As [my colleagues wrote](#) recently, Europe's relationship with natural gas has changed a lot in four years.

The bad news is things look much worse in percentage terms. And that is what matters more for inflation and the chance of rate hikes.

How the rise in energy prices compares to 2022

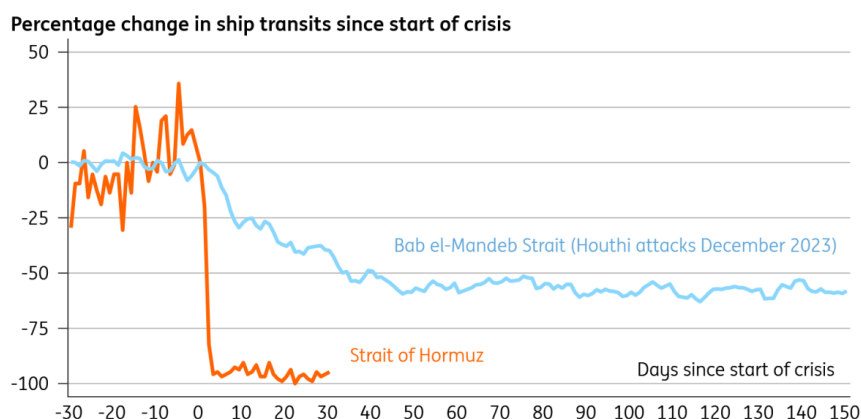


2 The economic impact is getting real

The other bad news is that there are scant signs of shipping recovering through the Strait of Hormuz. 50-60 tankers were typically transiting those waters daily in late February, according to IMF data. Now you can count them on one hand.

We've said all along that the economic impact of this crisis depends not just on how high energy prices rise, but how long they stay elevated. This week saw the first stories of widespread flight cancellations across Asia, together with other measures to conserve scarce fuel deliveries. This is a story that will likely hit Europe over the coming week or two, with increasingly unpredictable and potentially far-reaching economic effects.

How today's shipping disruption compares to the 2023 Red Sea crisis



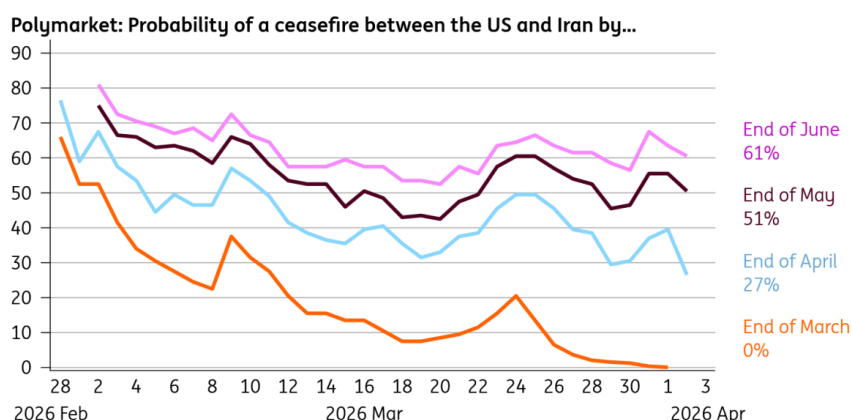
Source: Macrobond, ING

3 Markets are less confident about an April ceasefire

The starting point for the resumption of flows through the Strait is the question of a ceasefire. Betting markets think there's only a 27% chance that happens by the end of April, after another week of mixed messages from the US administration. Investors are weighing whether the White House will deploy the ground troops now stationed in the region.

Even if we do get a cessation of hostilities, does that guarantee the reopening of the Strait of Hormuz – and specifically, to what degree? And is that scenario enough to take oil prices sustainably back below 100 USD/bbl? And would that be enough to curtail the economic impact of the crisis and restore confidence in risk assets, or is it too late? Right now, there are no obvious answers.

What betting markets say about a ceasefire



Source: Macrobond

4 Markets still expect central banks to hike rates

"No obvious answers" could equally be applied to the central banks, though that's not how

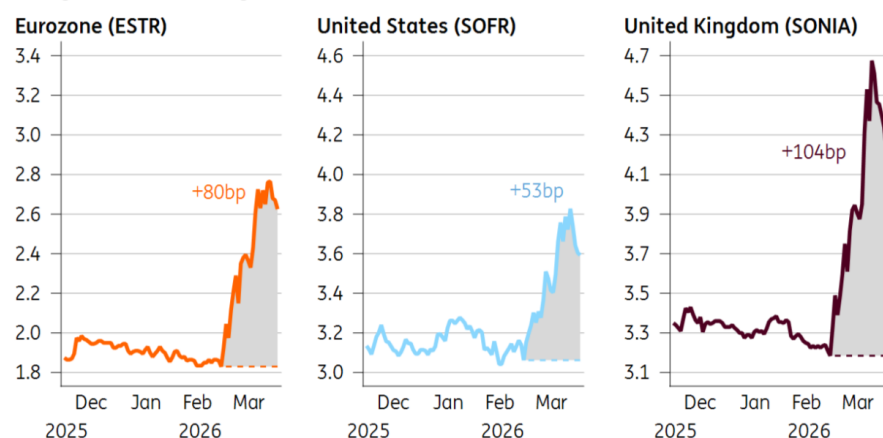
markets see it. In the eyes of investors, rate hikes are coming. Three are expected from the ECB by year-end and two from the Bank of England. The level of Bank Rate priced in a year's time is more than a percentage point higher in Britain than it was pre-crisis. Michiel Tukker, my Rates Strategy colleague, has a [neat chart](#) showing the clear relationship that's emerged between oil prices and central bank expectations since the war began.

Curiously, that relationship doesn't appear to have been jolted by what I felt was a pretty bold intervention by BoE Governor Andrew Bailey this week. In a very [rare comment](#) on market pricing, he said investors were "getting ahead of themselves". That felt like a not-so-subtle way of saying he doesn't intend to vote for a hike in April. And it's making me more comfortable in my current BoE call, which is that rates stay unchanged throughout this year.

How market interest rate expectations have evolved

Implied one-month rate in one year (1Y1M swap)

Change since 27 February



Source: Macrobond, ING

5 Governments aren't willing or able to splash the cash

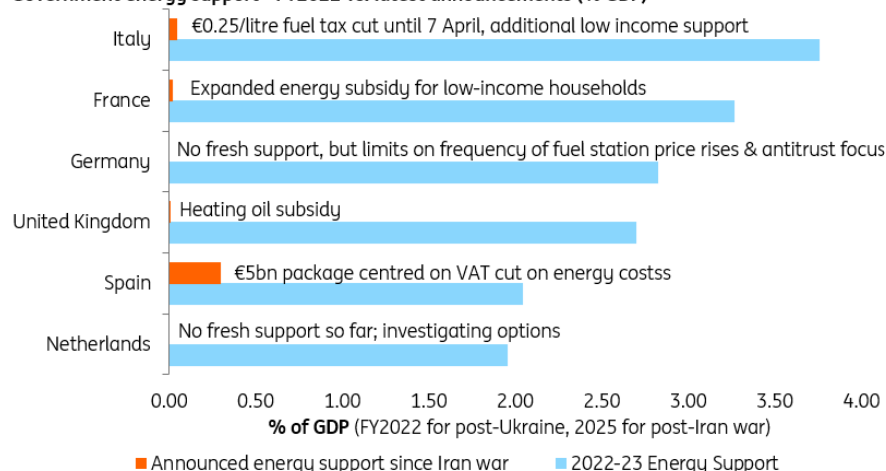
So why is he willing to go into battle with markets over the issue of rate hikes? One reason might be the lack of government intervention so far in this crisis. Back in 2022, the UK government unleashed support worth in excess of 2% of GDP, much of it universally applied. That support was one of the reasons the Bank of England opted to hike rates as far as they did.

This time, measures have so far totalled less than £100m. And it's a story that is mirrored across Europe. Partly it is a question of timing. The heating season is ending; the acute need for support isn't there yet. And the fact that gas prices are so far much more contained than in 2022 questions the need for widespread support, anyway.

Clearly, that could change if the crisis worsens and persists into the summer. But it is another reason for the ECB and BoE to hold off on lifting interest rates for the time being.

How energy support compares to FY2022

Government energy support - FY2022 vs. latest announcements (% GDP)



Source: OECD Energy Support Tracker, ING analysis of 2026 government policy

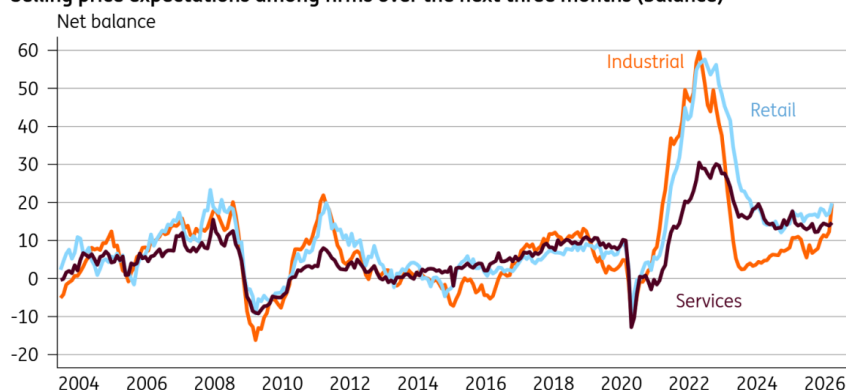
6 Firms don't have the pricing power they did four years ago

Another argument against hikes is that so far at least, there is scant evidence that firms are able to pass on higher energy costs to consumers. We saw this with last week's purchasing managers indices (PMIs), where the jump in the proportion of companies facing higher input costs was considerably larger than those who are now raising output prices. We saw something similar with the European Commission's survey this week; outside of industry, there isn't much sign of a rise in selling price expectations. Ditto the latest BoE survey of UK companies.

Sure, it's early days. And I still believe, as I [argued last week](#), that there is very little, if any data, due for release before the April meetings that can realistically influence the central banks' decisions. But the data fits with most of the evidence from recent months, which suggests this is not 2022 and firms lack the pricing power they had four years ago.

European selling price expectations

Selling price expectations among firms over the next three months (balance)

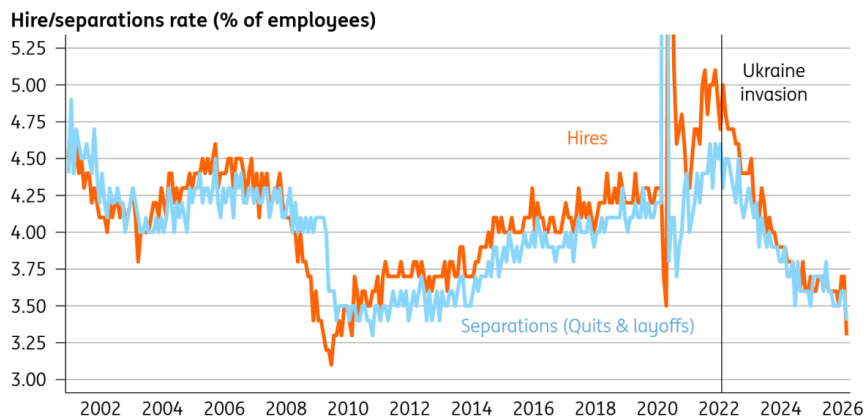


Source: Macrobond

Firms weren't willing to hire even before this crisis began

That lack of pricing power goes hand-in-hand with a jobs market that is in a much more fragile position than it was at the start of the Ukraine war. The US hiring rate (new hires as a share of total existing employees) fell again in February's data released this week and is now not dissimilar to where it was in the financial crisis. Something similar is true in Europe, particularly the UK. If anything, wage growth is more likely to fall rather than rise in the short to medium-term. That goes firmly against the idea that this crisis will lift inflation in the long-lasting way central banks worry about.

The US hiring rate is perilously low



Source: Macrobond

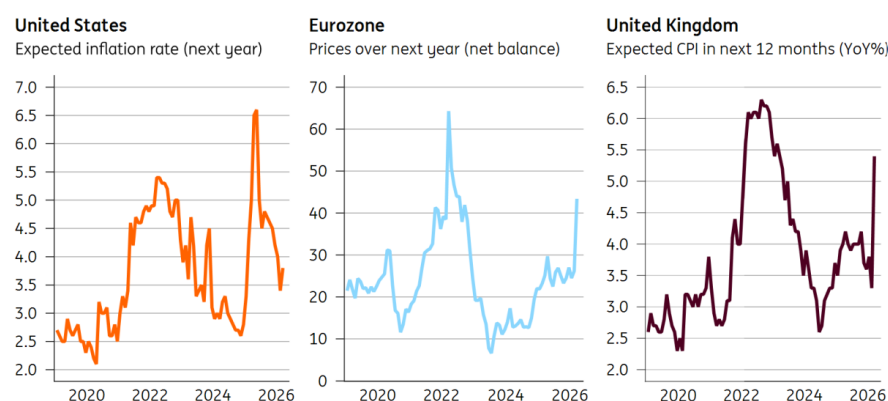
8 Surprise! Consumers are noticing higher prices

Having said all that, if there's one chart that gives me pause for thought, it's this. Consumer inflation expectations are rising - sharply. That is not remotely surprising, given these surveys essentially just track petrol prices. Nor, in a weaker jobs market, does it mean they have the power to respond by chasing higher wages.

But for central banks that worry more about the risk of inflation expectations becoming "unanchored", the clear risk is that they look at this chart and push ahead with rate rises to send a clear message. Fed Chair Jerome Powell said again this week that five years of above-target inflation may well start to filter more permanently into the consumer and business mindset.

I'm still not sold, as you might have gathered. And for the time being, our house view is unchanged. We don't expect rate hikes on either side of the Atlantic, at least when it comes to those April meetings that are drawing ever closer.

Consumer inflation expectations are rising



Source: Macrobond, ING

In the absence of March ECB consumer inflation expectations data, the European Commission survey looks at the net balance of consumers expecting prices to rise. US data from the University of Michigan. UK data from YouGov/Citi

THINK Ahead in developed markets

United States (James Knightley)

- **CPI (Fri):** March consumer price inflation will be the big report to watch next week with surging gasoline prices set to push annual inflation back above 3%. While natural gas prices have actually fallen in the US, limiting the potential impact from the Middle East conflict on broader utility bills, gasoline does have a weighting of more than 3% in the inflation basket. Add in lubricants, distribution costs and air fares, and we suspect that the month-on-month change in CPI could total 1%, lifting the year-on-year rate to 3.4% from 2.4%. After all, on a non-seasonally adjusted basis, the national average retail price of gasoline was \$3/gallon on 1 March versus more than \$4 at the beginning of April, with potential for it to hit \$4.50 in the next few weeks. Core inflation will be more muted, but still a relatively hot 0.3% or even 0.4%MoM.
- Nonetheless, we continue to doubt the Fed will raise interest rates. We think there is a greater chance that today's energy shock is demand destructive in that households have less discretionary spending power. This should diminish the chances of broader, more persistent inflation while the cooling jobs market remains a growing issue for the Federal Reserve.

THINK Ahead in Central and Eastern Europe

Poland (Adam Antoniak)

- **NBP rates (Wed):** A moderate increase in CPI inflation in March, the absence of any signs of upward price pressure apart from fuels and government intervention on retail petroleum prices should provide policymakers with enough comfort to stick to a wait-and-see approach for now. We expect rates to remain unchanged in April. The Monetary Policy Council (MPC) will closely monitor upcoming data searching for any signs of broader price pressure (core inflation, food prices) or second round effects. Our baseline scenario assumes the National Bank of Poland rates will stay at the current level of 3.75% by the end of this year.

Hungary (Peter Virovacz)

- **Industry/retail/inflation** (Wed): We are approaching an ultra-rare Super Wednesday, when the Hungarian Central Statistical Office will release three key sets of data. Starting with the data related to the real economy, we see strong retail sales figures for February, despite the expected drop in the fuel sub-sector due to extremely high fuel sales in January. After two consecutive months of growth, we expect the trend to end and industrial production to probably shrink on a monthly basis, with year-on-year stagnation. However, the most interesting data will be the March inflation figures. Despite the government acting quickly to cap fuel prices, the impact will still be significant. The increase in the minimum wage will probably continue to put pressure on service prices, and the significant weakening of the forint will also push up imported inflation. Nevertheless, thanks to the low starting point, inflation will appear acceptable at 2.2% year-on-year, but we anticipate further acceleration in the coming months.

Czech Republic (David Havrlant)

- **CPI** (Tue): Inflation in March likely quickened on the back of rising fuel prices, reflecting elevated global oil and natural gas prices along with a stronger dollar. The fuel reaction is likely to be swift, and other price segments are about to follow suit with a lag. February's real retail sales likely remained robust, supported by real wage gains, despite the registered unemployment rate being set to hover around recorded levels observed since 2017. Czech industry has stabilised in the previous year, and February's output is expected to carry on in annual growth, but still at a rather muted pace.

Key events in developed markets next week

Country	Time Data/event	ING	Prev.
Monday 6 April			
US	1500 Mar ISM Services PMI	53	56.1
Tuesday 7 April			
US	2000 Feb Consumer Credit Change (USD bn)	12	8
Eurozone	0900 Mar HCOB Services PMI Final	-	50.1
	0900 Mar HCOB Composite PMI Final	-	50.5
Germany	0855 Mar HCOB Services PMI Final	51.2	51.2
	0855 Mar HCOB Composite PMI Final	51.9	51.9
France	0850 Mar HCOB Composite PMI	-	48.3
UK	0930 Mar S&P Global Serv PMI Final	-	51.2
	0930 Mar S&P Global Composite PMI Final	-	51
Italy	0845 Mar Composite PMI	-	52.1
Spain	0815 Mar Services PMI	-	51.9
Wednesday 8 April			
US	1000 Mar 18 FOMC meeting minutes	-	-
Eurozone	1000 Feb Retail Sales (MoM%/YoY%)	-/-	-0.1/2
Germany	0700 Feb Industrial Orders (MoM%)	8	-11.1
France	0745 Feb Trade Balance (EUR bn)	-	-1.8
Switzerland	0800 Mar Unemployment Rate	-	3
Thursday 9 April			
US	1330 Feb Personal Income (MoM%)	0.4	0.4
	1330 Feb Personal Consumption (MoM%)	0.1	0.4
	1330 Feb Core PCE Price Index (MoM%)	0.4	0.3
	1330 Q4 GDP Final	0.7	0.7
	1330 Initial Jobless Claims (000s)	215	-
Germany	0700 Feb Industrial Output (MoM%/YoY%)	1.5/1.0	-0.5/-1.09
	0700 Feb Exports	2.5	-2.3
	0700 Feb Imports	4	-5.9
	0700 Feb Trade Balance (EUR bn)	18	21.2
Netherlands	0530 Mar CPI NSA (MoM%/YoY%)	2.7	2.4
Greece	1000 Feb Industrial Output (YoY%)	-	5.3
Friday 10 April			
US	1330 Mar Core CPI (MoM%/YoY%)	0.3/2.7	0.2/2.5
	1330 Mar CPI (MoM%/YoY%)	1.0/3.4	0.3/2.4
	1500 Apr Michigan Sentiment Flash	50	53.3
	1500 Apr Michigan Conditions Flash	55	55.8
	1500 Apr Michigan Expectations Flash	47	51.7
Germany	0700 Mar CPI Final (MoM%/YoY%)	1.2/2.7	0.2/1.9
Italy	0900 Feb Industrial Output (MoM%/YoY%)	-/-	-0.6/-0.6
Canada	1330 Mar Unemployment Rate	-	6.7
Norway	0700 Mar CPI (MoM%/YoY%)	-/-	0.6/2.7
Netherlands	0530 Feb Manufacturing Output (MoM%)	-	0.4

Source: Refinitiv, ING

Key events in EMEA next week

Country	Time (BST) Data/event	ING	Prev.
Tuesday 7 April			
Czech Rep	0800 Feb Trade Balance (CZK bn)	24.3	19.3
	0800 Feb Retail Sales (YoY%)	5.4	5.0
	0800 Mar CPI (MoM%/YoY%)	0.8/2.1	-0.1/1.4
Romania	1300 Rate Decision	6.50	6.50
Wednesday 8 April			
Czech Rep	0900 Feb Unemployment Rate	5.1	5.2
Hungary	0730 Feb Retail Sales (YoY%)	3.9	3.5
	0730 Feb Industrial Output (MoM%/YoY%)	-1.0/0.0	1.5/-2.5
	0730 Mar CPI (MoM%/YoY%)	0.8/2.2	0.1/1.4
	0730 Mar Core CPI (YoY%)	2.5	2.1
Thursday 9 April			
Poland	- Apr NBP Base Rate	3.75	3.75
Czech Rep	0800 Feb Industrial Output (YoY%, WDSA)	1.0	2.4
Ukraine	1400 Mar CPI (MoM%/YoY%)	-/-	1/7.6
Serbia	1100 Rate Decision	5.75	5.75
Friday 10 April			
Russia	1700 Mar CPI (MoM%/YoY%)	0.6/5.8	0.7/5.9
Hungary	1000 Mar Budget Balance (Monthly, HUF bn)	-800	-2139

Source: Refinitiv, ING

Author

James Smith

Developed Markets Economist, UK

james.smith@ing.com

James Knightley

Chief International Economist, US

james.knightley@ing.com

Adam Antoniuk

Senior Economist, Poland

adam.antoniuk@ing.pl

Peter Virovacz

Senior Economist, Hungary

peter.virovacz@ing.com

David Havrlant

Chief Economist, Czech Republic

420 770 321 486

david.havrlant@ing.com

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Belgium's inflation rises only slightly in March; wait a minute...

Belgian inflation increased only modestly, from 1.4% to 1.6% in March. Given the renewed surge in energy prices, that looks surprising at first glance. But this apparent stability is misleading: a sharper rise in inflation is likely in the coming months



Belgium's exposure to energy-price fluctuations remains high

High exposure to energy prices

Between 2021 and 2023, Belgium was among the European countries most affected by the inflation surge. Inflation peaked at 12.3% in October 2022 and stayed above 5% for 19 consecutive months. Several factors explain this outcome, but the jump in energy prices was a key driver. Energy accounts for a relatively high share of Belgium's consumer price index (around 10%), compared with most other European countries. In addition, the way gas and electricity prices are incorporated into the index makes inflation particularly sensitive to energy-price swings.

Gas and electricity bills enter the index using the acquisition-price principle: each month, the index reflects prices on newly signed energy contracts (fixed- or variable-priced), rather than the prices paid under existing contracts. Households that previously locked in fixed-price contracts are therefore insulated from short-term market moves, yet the index responds immediately, which can amplify the measured inflation response. A methodological change aligned with Eurostat

recommendations is under review, but it is not expected to be implemented before 2027.

A question of timing

Against this backdrop, it may seem surprising that inflation has not (yet) been more affected by the escalation in the Middle East and its impact on oil and gas markets. While inflation in fuel prices rose from -1.8% in February to 3.8% in March, other energy components (gas, electricity and heating oil) were broadly stable. As a result, the overall energy component still made a negative contribution to headline inflation in March (-0.2 percentage points).

Once again, methodology largely explains the disconnect:

- Heating oil prices are smoothed over 12 months, reflecting the fact that households do not refill their tanks every month. A sharp increase in heating oil prices is therefore incorporated into the index only gradually.
- For gas and electricity, the limited impact mainly reflects timing. Reference prices for new contracts were updated at the beginning of March, before the Middle East escalation translated clearly into retail contract pricing. Later in the month, some suppliers attempted to raise reference prices for new contracts or temporarily stopped offering new fixed-price deals, but these changes were not captured in March's inflation calculation. A more visible increase in gas and electricity bills is therefore more likely to show up from April onwards.

Other disinflationary forces

It is also worth noting that several major categories put downward pressure on inflation in March, further masking the initial spillovers from higher energy costs to other goods and services. Most notably, food prices fell by 1.2% year-on-year in March, after rising by 0.3% in February.

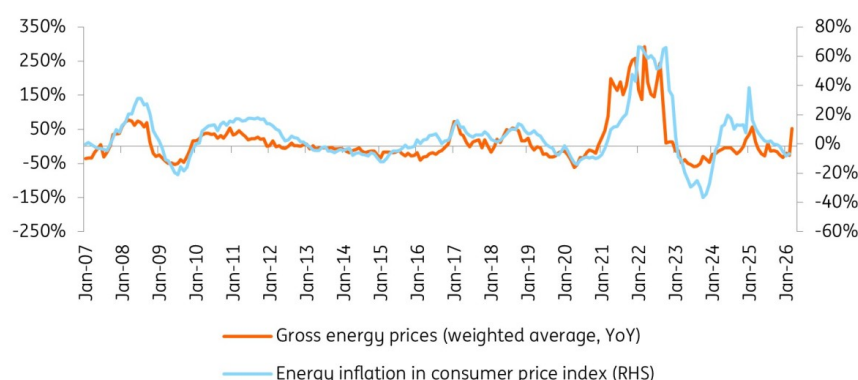
The wave is coming

Belgian inflation's strong sensitivity to energy prices has not disappeared: it has merely been delayed. This is all the more true because, so far, only limited measures have been taken to shield households from higher energy bills. The chart below illustrates the link between gross energy prices on international markets and energy inflation for final consumers. It compares (i) the year-on-year change in a synthetic index of gross energy prices in Belgium (22% gas, 36% electricity and 42% oil) with (ii) year-on-year inflation in consumer energy prices as captured in headline inflation.

Based on the latest available data, the synthetic gross energy price index has already risen by at least 50%. If the historical relationship holds, this could translate relatively quickly into roughly 13% inflation in the energy component of the consumer price index. Given its weight (8.4%) in the index, that would add about 1.1 percentage points to headline inflation, which was still slightly negative in March. It would therefore not be surprising to see inflation rise from 1.6% in March to close to 3% in April. And because the shock to energy bills is likely to persist and gradually spill over to other goods and services, inflation in Belgium could move toward 4% in the coming months.

Energy prices and inflation

Rising gross energy prices (gas, electricity and oil) should quickly translate into higher inflation in the energy component of the price index.



Source: Statbel, LSEG Datastream

A shock that is hard to pin down

In short, Belgium's exposure to energy price fluctuations remains high and should become more apparent from April onwards. Beyond that, inflation will depend on how the Middle East conflict evolves and on its effects on the production and transport of energy and other raw materials. Higher energy prices will also feed into wages through Belgium's automatic indexation mechanism. While a recent government measure will limit indexation for high salaries, further price increases cannot be ruled out as wage costs rise. Belgian inflationary pressures may therefore prove more important than the March reading suggests.

Author

Philippe Ledent

Senior Economist, Belgium, Luxembourg

philippe.ledent@ing.com

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Rates Spark: Markets don't see a solution yet

Trump's rhetoric does not offer markets a tangible solution. This will keep both oil prices up and risk aversion raised as we enter a long weekend. Despite pushback from the Bank of England, sterling rates continue to take a very hawkish stance compared to peers, which should start to weigh on the belly of the curve if materialised



Sterling rates are showing a higher sensitivity to oil prices than euro and dollar rates

Start of a long weekend will keep markets on edge

The hope was that US President Donald Trump would appease markets with concrete steps towards ending the war in the Middle East. Unfortunately, the rhetoric appeared largely unchanged, keeping markets on edge. Today is the last day before the start of a long weekend, with US markets closed on Friday; in Europe, Monday will also be a bank holiday. As such, we doubt oil can move materially lower, and risk sentiment should stay fragile.

In effect, that means the upward pressure on the front end of the curve will stay. The 5Y and 10Y points might face less upward pressure as growth concerns over the medium term intensify. Inflation markets continue to be positioned for a short-lived inflation spike, which then begs the

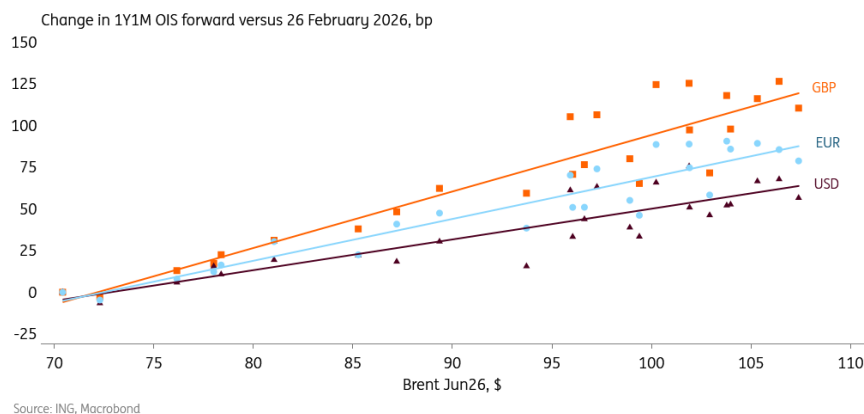
question of how much easing central banks follow up with after their hiking cycle. This will all depend on the resilience of the economy after tightening policy while facing higher energy costs. In the worst outcome, central banks might have to cut aggressively into accommodative territory to mitigate recession risks. Such a scenario helps explain why the belly of the curve can outperform in the current environment.

Markets are set on a hawkish Bank of England

Sterling rates are showing a higher sensitivity to oil prices than EUR and USD rates, and even dovish pushback from the Bank of England is achieving little. Our screen now shows almost 50bp of hikes being priced in for this year, which is a complete turnaround from the two cuts being priced in February. For every \$10 increase in the oil price, GBP rates are adding 34bp of tightening over the next year. This is just 25bp for euro rates and 18bp for dollar rates. Even BoE Governor Andrew Bailey didn't meaningfully move the needle to a more dovish stance despite being very vocal that "markets are ahead of themselves".

The hawkish reaction from markets is mainly a product of the hotter inflation outlook, reflected by a 2Y inflation swap of well above 4%. In comparison, this is just 2.7% for eurozone inflation. From a curve perspective, we would argue that a more aggressive hiking cycle would worsen the medium-term economic outlook more, helping the belly of the curve to outperform. These dynamics are already pronounced in the GBP curve, which has seen the 2s5s10s move from around -8bp to -15bp. If we do get a BoE as hawkish as markets are pricing in, then we can expect the 5Y point to see more downward pressure.

GBP rates are most sensitive to oil price moves



Thursday's events and market views

Markets will be closed on Friday, and this means investors will need to decide how much risk to take into the long weekend, which will also include a US payrolls report. But before that, we first have the Challenger jobs report and initial jobless claims on Thursday. The focus is still on the Middle East conflict, which means macro data is still secondary in terms of market drivers. More interesting could be the various central bank speakers, including the ECB's Francois Villeroy discussing monetary policy in Paris.

In terms of supply, we have France with 9Y, 10Y and 22Y OATs together with an 18Y green OAT for a total of €12.5bn.

Author

Michiel Tukker

Senior UK & Eurozone Rates Strategist

michiel.tukker@ing.com

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FX Daily: Back to escalation

We have seen the first part of the de-escalation trade in FX. The second – another USD leg lower – requires greater clarity on Strait of Hormuz reopening plans, which currently appear distant after the headlines overnight. A new wave of escalation is now driving EUR/USD down again. Any downside surprise in tomorrow's US data could support more Fed repricing



Another USD leg lower requires greater clarity on Strait of Hormuz re-opening plans, which appear more distant once again after Trump's latest address to the nation

The next FX Daily will be published on Tuesday 7 April

USD: Return to risk-off sentiment supports the US dollar again

USD bearish trades gathered further momentum yesterday on optimism around a potential Middle East de-escalation. However, it didn't get far after a new wave of escalation in US President Donald Trump's speeches to the nation indicated more weeks of fighting, returning a risk-off sentiment to the market today and reversing most of the recovery from the last two days.

President Trump said Iran has requested a ceasefire, though he stressed this would be contingent on the reopening of the Strait of Hormuz. This marks a shift in Trump's narrative relative to recent days, when he had suggested a willingness to distance the US from decisions surrounding the

Strait's fate.

We'll hear from the Federal Reserve's Lorie Logan and Michelle Bowman today, but Monday's dovish-leaning comments by Chair Jerome Powell will continue resonating loudly with markets. Tomorrow's non-farm payrolls can have decent impact potential. They won't be telling us too much about the war's impact on jobs yet, but they could shed more light on the state of the labour market amid the shock, which could materially steer the Fed's assessment at this stage.

Yesterday's ADP payrolls were stronger than expected at 62k, and the ISM manufacturing employment index was almost unchanged at 48.7 in March. Consensus for the NFP is 65k, our macro team's call is 60k and the Bloomberg whisper number is now 40k. We agree with consensus that unemployment will remain at 4.4% – any material jump here can have an outsized impact.

Watch for reduced liquidity conditions tomorrow and Monday due to the Easter holidays.

Francesco Pesole

📉 EUR: Back to 1.150 levels on a wave of lost hope

With the conflict re-escalating and oil prices well above \$100/bbl again, EUR/USD sees another dip back well above 1.160 and reaching 1.150 this morning, not far from local lows.

It's worth noticing that CFTC data suggests a neutral speculative positioning on EUR/USD as of 24 March, as opposed to stretched net-longs earlier in March. Other things equal, it suggests better sustainability of a EUR rally if we see any de-escalation over the weekend.

Ultimately – as stressed [yesterday](#) – the European Central Bank's stance during this de-escalation phase remains key for the euro. Markets are tempted to unwind tightening bets further, but the ECB's hawkish rhetoric may prevent pricing from falling below two hikes (now 67bp) unless policymakers signal some flexibility towards a more dovish rhetoric. If that doesn't happen, the euro could emerge as an outperformer in the coming days.

Francesco Pesole

📉 GBP: A rate protest from Andrew Bailey

Bank of England Governor Andrew Bailey appeared to deliver a rate protest in an exclusive interview with Reuters yesterday. He emphasised that markets were 'getting ahead of themselves' in pricing a series of rate hikes this year. The surge in short-dated UK rates during this Middle East crisis, fuelled by what was perceived as a very hawkish MPC meeting on 19th March, has clearly damaged both business and consumer confidence. A key takeaway from yesterday's interview was that the BoE should fulfil its remit in a way that causes the least damage to the economy and the people. We saw a more than 100bp rise in two-year swap rates last month, and the BoE's role in that spike is clearly an issue.

Whether Governor Bailey is successful in putting the hawkish genie back in the bottle will depend on the data. Today sees the release of the important BoE Decision Maker Panel (DMP) survey – a survey of 2,000-2,500 CFOs. Of key interest will be what CFOs are thinking about pricing power. Input costs will have gone up, of course. But will CFOs feel that selling prices and wage costs will rise too?

We are in Andrew Bailey's camp with the view that a growing output gap and weak pricing power mean that there are limited chances of second-round effects from this energy supply shock. Were the DMP to confirm this, with, say, expected selling prices remaining near 3.0% year-on-year and wage costs at 3.5/4.0% YoY, we suspect the market can further rein in the near 50bp of expected BoE tightening this year. If so, EUR/GBP could make its way back to the 0.8790/8800 area – where it was trading at the end of February.

Chris Turner

⬇️ CEE: Far from relief

The region is preparing for a long weekend, with the Czech Republic and Hungary closed for trading on Friday. Given the uncertain geopolitical environment, this should lead to a reduction of risk ahead of the extended week despite the risk-on sentiment in the last two days.

The calendar doesn't have to offer today, except for a government meeting in the Czech Republic, which may see a discussion of potential measures to protect households from higher fuel prices, with a margin freeze on the table.

Turkey will release its March inflation on Friday. We expect a slowdown from 3.0% to 2.2% month-on-month, but still a higher reading than before the fuel shock, resulting in an increase from 31.5% to 32.2% YoY. This should confirm that Central Bank of Turkey rate cuts are not on the table at this time – at least for the next few months, until more clarity is seen on the impact of oil prices and the development of domestic inflationary pressures.

CEE FX saw some relief yesterday as the zloty and koruna joined the forint rally, but as global headlines suggest, we are still far from any visible relief. EUR/PLN and EUR/CZK remain close to their local highs, and we can expect some reversal in the recent EUR/HUF slide of the past few days. Getting back to 390 could be too far.

Frantisek Taborsky

Author

Frantisek Taborsky

EMEA FX & FI Strategist

frantisek.taborsky@ing.com

Francesco Pesole

FX Strategist

francesco.pesole@ing.com

Chris Turner

Global Head of Markets and Regional Head of Research for UK & CEE

chris.turner@ing.com

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The Commodities Feed: Oil jumps amid Trump's escalation warning

Oil rebounded sharply on Thursday, rising more than 5% after two days of declines, as US President Donald Trump threatened a further escalation of the war with Iran, injecting fresh uncertainty into energy markets



Oil prices have surged on the back of US President Donald Trump's latest televised speech from the White House

Energy – Oil jumps after Trump's war escalation threats

Oil rebounded sharply on Thursday, rising more than 5% after two days of declines, as US President Donald Trump threatened a further escalation of the war with Iran, injecting fresh uncertainty into energy markets.

Brent was trading above \$107/bbl, while WTI was close to \$106/bbl on Thursday morning following Trump's address to the nation last night. The President said the US would hit Iran "extremely hard" over the next two to three weeks and could target "each and every one" of the country's power plants. He also claimed the Strait of Hormuz would reopen "naturally" once the conflict ends, though he offered no details or a clear timeline. Trump additionally urged allies reliant on Middle Eastern energy to help resolve the near closure of Hormuz.

Even if shipping through the Strait of Hormuz resumes, a return to pre war market conditions is

likely to be slow, as upstream production restarts, logistics normalisation and inventory rebuilding will take time.

US inventory data were mixed. EIA figures showed crude stocks rose for a sixth consecutive week, increasing by 5.5m barrels, below the 10.3m barrel build reported earlier by the API. Total crude inventories climbed to 461.6m barrels, the highest since June 2023 and close to the five year average. Cushing stocks rose by 520k barrels to 31.5m barrels, also the highest since July 2024. Crude imports were steady at 6.5m b/d, while exports increased by 199k b/d to 3.5m b/d.

In refined products, gasoline and distillate stocks fell by 0.6m barrels and 2.1m barrels, respectively, reflecting weak refinery runs despite rising implied demand amid heightened supply concerns. Refinery utilisation slipped 0.8ppt week-on-week to 92.1%.

Metals – Aluminium rallies on more Gulf smelter halts

Aluminium prices rallied above \$3,500/t on Thursday after reports that Emirates Global Aluminium (EGA) halted operations at its Al Taweelah smelter after the site was hit by Iranian missiles and drones over the weekend, according to Wood Mackenzie. According to the report, utilisation at Alba's smelter has fallen to around 30%, from 81% previously. For now, these reports have not been confirmed by the companies.

If confirmed, a halt at EGA's 1.6 mtpa Al Taweelah smelter, Alba's reduced operations and earlier curtailments at Qatalum would take around 3 mtpa of capacity offline – close to half of Middle East aluminium production. That marks a sharp escalation from the roughly 8-9% disruption we were flagging in mid March.

This would imply materially deeper aluminium deficits [across all our scenarios](#), lifting the severe disruption case to around 2-2.5 Mt, with even the base case now pointing to a deficit of roughly 1 Mt.

With Chinese supply capped, Indonesia's ramping up constrained by power and limited capacity growth elsewhere, aluminium's fundamentals remain firmly supportive. Prolonged Middle East disruptions add further tightening risk.

EGA has also reportedly moved to sell large volumes of alumina in the wake of the strike, according to a Bloomberg report.

LME aluminium prices are up around 17% year to date, with the forward curve in steep backwardation. That said, escalating global growth concerns could still lead to some demand destruction, partially offsetting the supply shock.

Aluminium, along with other base metals, pared earlier gains on Thursday morning after President Trump reiterated threats to attack Iran's power plants, with rising rate risks renewing concerns over demand destruction.

Agriculture – Sugar prices ease on weaker demand signals

Sugar prices fell for a fourth consecutive session, briefly dipping towards US¢15/lb (-3% DoD), as softer oil prices reduced incentives for mills to divert output towards ethanol. While the front month contract is still up almost 12% over the past month – its strongest run since 2023, supported by crude's earlier rally on Middle East tensions – recent hopes of de escalation have

weighed on energy markets and, in turn, on sugar. Elevated volatility and uncertainty around the timing of any resolution have also encouraged position trimming. Meanwhile, favourable weather conditions in Brazil continue to support cane development, reinforcing expectations of ample supply in the upcoming season.

In India, data from the National Federation of Cooperative Sugar Factories show 2025/26 sugar production up 9% year-on-year to 27.1mt as of 31 March. The number of operating mills fell to 74, from 113 a year earlier, while cumulative cane crushing rose to 283.5mt, up from 265.5mt over the same period.

Author

Ewa Manthey

Commodities Strategist

ewa.manthey@ing.com

Warren Patterson

Head of Commodities Strategy

Warren.Patterson@ing.com

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Asia week ahead: Korea rate decision, inflation data from China and Taiwan

The Bank of Korea is likely to leave interest rates unchanged, while China and Taiwan release inflation data amid surging global oil prices



Asia Research highlights of the week

[South Korea's economy benefits from robust chip exports and fiscal support](#)

[Japanese business sentiment remains solid, supporting April BoJ hike](#)

[Oil drives South Korean inflation higher, but government curbs keep it contained](#)

[Slower Tokyo inflation unlikely to deter Bank of Japan's April hike](#)

[South Korean data points to firm first-quarter growth, but Iran risks abound](#)

[Chinese PMI rebounds as Iran impacts start trickling through](#)

South Korea: BoK expected to keep rates steady

The Bank of Korea is expected to leave policy rates unchanged on Friday. Strong chip demand has helped offset the effects of commodity supply constraints. Oil prices are driving inflation higher, but government measures are keeping it contained. Against this backdrop, the BoK is likely to take

a wait-and-see approach for now to analyse the impact of Middle East shocks. We expect price pressures to increase and financial imbalances to worsen in the coming months. This could prompt the BoK to hike rates in the third quarter.

Taiwan: Inflation to moderate, but underlying pressures remain

Taiwan's March inflation data is out on Wednesday. We expect inflation to moderate slightly to 1.6% year-on-year, lower than February's Lunar New Year holiday-boosted 1.75% level. The data is likely to show early signs of price pressures emerging in the energy sector. With inflation risks rising and growth consistently beating forecasts, the odds are shifting toward the central bank's next move being a hike rather than a cut. Unless inflation significantly overshoots expectations in the coming months, policymakers will likely stay on hold when they next meet in June.

China: Inflation data to show some momentum

China releases its March inflation data on Friday. We expect that the positive price momentum from the last couple of months carried over into March, especially with the surge in energy prices amid the Iran War. The price subcomponents of the purchasing managers' index data, which reached their highest monthly levels since 2022, suggest inflation will heat up in March. We expect CPI inflation to remain elevated at 1.2% YoY and PPI inflation to return to positive territory for the first time since 2022, at 0.6%. Markets will also monitor local gasoline price moves, as the bi-weekly adjustment is expected on 7 April. Prices have climbed since the outbreak of the war, but at a much slower pace than overall crude oil prices.

Key events in Asia next week

Country	Time (GMT+8)	Data/event	ING	Prev.
Monday 6 April				
India	1300 Mar	S&P Global Services PMI Final	-	57,2
Singapore	1300 Feb	Retail Sales (MoM%/YoY%)	-/-	6.1/-0.4
Tuesday 7 April				
China	1600 Mar	FX Reserves (USD bn)	-	3,4
Japan	0730 Feb	All Household Spending (MoM%/YoY%)	3.0/-0.2	-2.5/-1
Philippines	0900 Mar	CPI (MoM%/YoY%)	-/-	0.2/2.4
	1600 Feb	Budget Balance (PHP bn)	-	165,4
Taiwan	1620 Mar	FX Reserve (USD bn)	-	605,5
Singapore	1700 Mar	Foreign Reserves (SGD bn)	-	416,1
Wednesday 8 April				
Japan	0730 Feb	Labour Cash Earnings (YoY%)	2,9	3
India	1230	Rate Decision	-	5,25
South Korea	0700 Feb	Current Account Balance (USD bn)	-	13,3
Indonesia	1100 Mar	Forex Reserves	-	151,9
Taiwan	1600 Mar	CPI (MoM%/YoY%)	1,6	1,8
Thursday 9 April				
Japan	1300 Mar	Consumer Confidence	38	40
Friday 10 April				
China	0930 Mar	CPI (MoM%/YoY%)	1,2	1,3
	0930 Mar	PPI (YoY%)	0,6	-0,9
Japan	0750 Mar	PPI (MoM%/YoY%)	0.5/2.2	-0.1/2
Indonesia	1100 Mar	Consumer Confidence Index	-	125,2
Taiwan	1600 Mar	Imports	-	6,8
	1600 Mar	Exports	-	20,6
	1600 Mar	Trade Balance (USD bn)	-	12,8
South Korea	0900	Rate Decision	2,5	2,50

Author

Lynn Song

Chief Economist, Greater China

lynn.song@ing.com

Min Joo Kang

Senior Economist, South Korea and Japan

min.joo.kang@ing.com

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The Commodities Feed: Oil falls below \$100 on optimism over Iran war

Oil prices fell after US President Trump signalled a potential end to the war with Iran. Brent slipped below \$100/bbl on Wednesday morning, while WTI also fell below \$100, after Trump said the US could leave Iran within two to three weeks, suggesting an agreement with Tehran may be reached but is not required for the conflict to end



Oil prices have fallen back below \$100 a barrel

Energy – Oil falls on optimism over Iran war

Oil prices fell after US President Trump signalled a potential end to the war with Iran. Brent slipped to below \$100/bbl on Wednesday morning and WTI also traded below \$100, after Trump said the US could leave Iran within two to three weeks, suggesting an agreement with Tehran may be reached but is not required for the conflict to end. The President is due to address the nation at 9pm ET today with an update on Iran.

The *Wall Street Journal* also reported that the UAE has urged the US and European and Asian military powers to form a coalition to open the Strait of Hormuz by force.

Meanwhile, China and Pakistan jointly called for an immediate ceasefire and for shipping through Hormuz to be safeguarded.

On Tuesday, Trump warned allies to buy US jet fuel or risk losing access via the Strait, telling countries reliant on oil to “start learning how to fight for yourself.”

Even if the Strait reopens, clearing the vessel backlog would take time, with production, exports and LNG flows normalising only gradually rather than immediately.

Meanwhile, Saudi Aramco may raise the official selling price of Arab Light for Asian buyers next month, while key US crude grades are trading at their strongest premiums since the Covid 19 period.

In US inventory data, API figures showed crude stocks rose by 10.3mb last week, far above expectations for a 1.3mb build. Gasoline and distillate inventories fell by 3.2mb and 1.04mb, respectively, which should continue to support product cracks. The EIA report is due later today.

In gas markets, European prices fell sharply, with TTF down more than 7.7% on Tuesday amid favourable weather forecasts and hopes of de-escalation in the Iran conflict. Strong wind generation could weigh on gas demand for power, while weaker industrial demand and a lighter maintenance season in Norway add to the bearish tone. EU gas storage levels have fallen to 28% as of 30 March, well below the five-year average of 41%, prompting the Energy Commissioner to urge early injections to avoid a late season supply crunch.

Metals – Gold extends gains for third session

Gold extended gains for a third session on Wednesday on hopes that the war in the Middle East may be nearing an end. Spot prices climbed above \$4,700/oz, while equities rallied and the US dollar fell, after President Trump said he expected the US to end the war within weeks.

Also, earlier this week, Fed Chair Jerome Powell said long term US inflation expectations remain anchored and policy is “in a good place to wait and see”.

Still, despite the rebound this week, gold’s almost 12% decline in March was its worst monthly performance since October 2008.

Gold remains vulnerable to a broader liquidity squeeze and a firmer US dollar, though so far pullbacks have been met with buying rather than a loss of confidence.

Upcoming data on central bank purchases will be key to gauging whether the early year slowdown in official sector buying marks a more sustained pause or merely a temporary breather, with recent Turkish sales yet to be fully reflected.

In base metals, on the supply side, Chile posted its lowest monthly copper output in almost nine years, underscoring persistent structural constraints. February production fell to 378,554 tonnes, down 8.5% m/m and 4.8% y/y, according to INE data, as declining ore grades and underperformance at key mines weighed on output. The last time production was this low was March 2017, when strikes halted operations at BHP’s Escondida mine. Chile, which accounts for around a quarter of global mine supply, has now seen output decline on an annual basis for seven consecutive months.

Elsewhere, Ivanhoe trimmed production guidance for the Kamoā Kakula mine, lowering its 2026 outlook to 290-330kt from 380-420kt and cutting its 2027 forecast to 380-420kt from 500-540kt, as it adopts a more conservative mining plan following last year’s flooding.

According to the latest LME COTR report, speculators increased net long positions in aluminium by 1,285 lots in the week ending 27 March, snapping six consecutive weeks of declines and lifting total net bullish bets to 82,987 lots. The increase reflected fresh long positions and short covering amid tight global supply. In contrast, net longs in copper fell by 2,302 lots to 38,729, while zinc net longs declined by 4,393 lots to 40,171 after two weeks of gains.

Agriculture – Cocoa rises on adverse weather

Cocoa prices in the US and London jumped more than 5% amid concerns that El Niño could further disrupt production in key growing regions. Over half of Côte d'Ivoire and nearly two-thirds of Ghana are experiencing drought conditions, threatening cocoa formation and potentially slowing the mid crop harvest. Risks could intensify later this year if El Niño develops, bringing additional dryness to West Africa during a critical stage of the next main crop.

The USDA's Prospective Plantings report points to lower US corn and wheat acreage this year, while soybean plantings are set to rise. Corn acreage is forecast to fall to 95.3m acres from 98.75m in 2025, above market expectations of 94.5m acres, reflecting a shift toward soybeans after last year's record harvest. Wheat acreage is projected at 43.8m acres, the lowest since 1919, down 3% YoY and below expectations, pressured by high input costs, fertiliser shortages and weaker profitability. Soybean acreage is expected to increase 4% YoY to 84.7m acres, slightly below market expectations despite ongoing trade frictions with China.

The USDA's quarterly stocks report showed US corn inventories at 9.02bn bushels as of 1 March, up 11% YoY but below expectations of 9.1bn bushels. Soybean stocks rose 10% YoY to 2.1bn bushels, marginally above expectations, reflecting slower export demand earlier in the season. Wheat inventories stood at 1.3bn bushels, up 5% YoY and broadly in line with forecasts.

Elsewhere, Uganda's coffee exports rose 15% YoY (+15% MoM) to 651.9k bags in February, supported by higher output. Cumulative shipments for the 2025/26 season (Oct-Feb) reached 3.05m bags, up 26% YoY.

Author

Ewa Manthey

Commodities Strategist

ewa.manthey@ing.com

Warren Patterson

Head of Commodities Strategy

Warren.Patterson@ing.com

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FX Daily: ECB hawks can make euro a winner in de-escalation trade

The dollar has taken a hit on a new de-escalation trade. President Trump has now signalled that the war will end in two to three weeks, but plans for the reopening of the Strait of Hormuz remain quite unclear. EUR front-end rates are catching up with the downside today, but hawkish ECB comments can make them stickier and make the euro emerge as an outperformer



Could the euro emerge as an outperformer?

📉 USD: Hormuz remains key for more downside

Markets are making another attempt at playing the de-escalation trade. The trigger was reports that Iranian officials are leaning towards dialogue and Trump explicitly saying the US will end the war within two to three weeks. Many unanswered questions remain on the Strait of Hormuz, and despite a decline, Brent remains above \$100/bbl.

Trump is insisting Hormuz is not a US concern and should be up to other NATO countries to reopen it. At the same time, he hinted that once the US leaves the area, the Strait should reopen automatically, and the *WSJ* reported that the UAE is lobbying for a military coalition to end the blockage.

We think some better clarity and timeline on a reopening of the Strait is a necessary condition for DXY to trade back to the 99.0, 23 March lows. That said, the bias seems to be on the downside today.

The highlight in the US calendar is ADP payrolls, expected at 40k. Any soft print here may still be telling us more about fragility in the jobs market into the oil shock rather than already showing a war impact on employment. Either way, it would endorse the ongoing dovish repricing in the Fed curve, which is now attempting to price back in one cut (-11bp by December).

Francesco Pesole

EUR: Watch the pricing resilience

One key theme for the euro moving on is how flexible on the downside ECB rate expectations will be. That's because, as mentioned, markets appear comfortable pricing back Fed easing. After all, the Fed has a dual mandate, and its members have not been too hawkish since the start of the war.

The ECB has only an inflation mandate and its officials have sounded more convincingly hawkish so far. Yesterday, the most dovish Governing Council member, Fabio Panetta, openly discussed preventing a wage-price spiral, and while warning policy action should remain proportionate, the fact that he discusses 'action' as a base case sounds like a nod to hike expectations.

These comments raise the chances that EUR front-end rates will prove sticky on the way down, as markets may require some dovish comments from the ECB to price out hikes even if oil prices keep declining. This morning, markets have abruptly trimmed bets on ECB tightening to 63bp, but there was much more resilience in those hawkish bets yesterday.

A scenario where markets find it much easier to reprice the USD swap curve lower relative to the EUR one in an oil selloff significantly raises the upside potential for EUR/USD. That also increases the euro's potential against other currencies where the rate expectations are more vulnerable to a dovish repricing (like SEK, GBP) or more strictly exposed to oil prices (CAD, NOK)

Since our base case remains no ECB action and two Fed cuts, we remain cautious with our EUR/USD call at 1.160 for the end of 2Q.

Francesco Pesole

GBP: Rates can drop faster than in the eurozone

EUR/GBP faced upside volatility on the back of a tightening rate differential yesterday, but is coming back a bit lower after this morning's drop in ECB pricing.

Aside from Bank of England officials sounding a bit less hawkish in off-meeting comments, two notions can help the dovish repricing be deeper in the UK than the eurozone. First, the BoE was ready to cut before the war started. Second, the growth impact on the UK from the energy shock is expected to be the largest among OECD countries – a dovish argument.

We retain a bias to the upside for EUR/GBP on the back of that, and our expectations that the whole of BoE tightening will also be priced out. In our view, 0.880 remains a realistic target.

Francesco Pesole

CEE: Polish inflation suggests weaker impact of fuel prices on inflation in the region

The first inflation figures for March from Poland indicate a weaker-than-expected impact of high global energy prices. Inflation **jumped** from 2.1% to 3.0% YoY, but below market expectations. The main and essentially only driver was fuel prices, similar to the eurozone. Fuel prices in Poland jumped by 15.4% MoM, suggesting that elsewhere in the region, especially the Czech Republic, a market most similar to it, without caps, could also be weaker next week.

Today, PMIs for March will be published across the region, which should not yet be fully affected by the start of the US-Iran conflict, given that the survey is conducted around the middle of the month, when it was not yet clear that the conflict would last for a long time. However, the risk here is clearly to the downside compared to our and market expectations.

Yesterday's risk-on sentiment pushed rates down in the CEE region along with core markets, however in FX markets we saw relief only in the case of the forint, which remains the most beta currency in the region. On the other hand, the zloty and the koruna are not seeing much market favour for now. They probably require a longer period of relief and a clearer signal in the global story, given that the previous sell-off was not as severe.

Frantisek Taborsky

Author

Francesco Pesole

FX Strategist

francesco.pesole@ing.com

Frantisek Taborsky

EMEA FX & FI Strategist

frantisek.taborsky@ing.com

Chris Turner

Global Head of Markets and Regional Head of Research for UK & CEE

chris.turner@ing.com

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Eurozone inflation surges to 2.5% as Middle East shock starts to bite

The increase from 1.9% to 2.5% was entirely due to higher energy prices, with core inflation and food prices still falling. But the longer the shock lasts, the higher the risk of second-round effects causing broader elevated inflation



A litre of standard Euro-95 petrol has surged in recent weeks across Europe

The ECB is no longer in its 'good place'

Say goodbye to the ECB's good place. After a long period of eerily stable inflation despite global disruption, eurozone inflation has once again shot up thanks to the surge in energy prices. The price at the pump is the main culprit, with a litre of Euro-95 up almost 15% in the past month.

The other main categories of inflation didn't show an impact so far. Food inflation dropped from 2.5 to 2.4%, and core inflation fell from 2.4 to 2.3%. Both goods and services inflation moderated, indicating that price effects outside of energy were quite benign.

A rough ride for consumers

But looking ahead, you cannot see the energy price increase in isolation. It's all about the Middle East, which dominates the inflation outlook, and not just when it comes to energy prices, but also

expect upside risk to food and goods prices given fertiliser shortages and broader supply chain problems stemming from the war. Businesses in the industry sector just increased their selling price expectations to the highest level since early 2023, for example.

And consumers expect another rough ride, the past shock still fresh in memory. Inflation expectations just increased to levels only seen in the early 1990s and during the first half of 2022. For the European Central Bank, concerns about anchoring inflation expectations around 2% will have high priority at this stage. The longer the disruption lasts, the greater the likelihood of broader increases in headline and core inflation. With much uncertainty around how the Middle East conflict will evolve, many scenarios for inflation remain possible, and that's why the ECB is right to be on high alert.

Author

Bert Colijn

Chief Economist, Netherlands

bert.colijn@ing.com

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Watch: European pharma's fight to stay relevant

What does Europe need to do to stop its pharmaceutical industry from becoming 'yesterday's news'? ING's Diederik Stadig says the sector still has trump cards to play, but it needs to act fast to avoid being squeezed out by US and Chinese competition. Read more about Diederik's thinking [here](#)



European pharma's fight to stay relevant

Europe's pharmaceutical industry needs to up its game if it's going to continue competing effectively with the likes of China and the US. So says ING's Senior Economist for Healthcare and Technology, Diederik Stadig. He points out that in 1990, nearly half of all global Research & Development spending took place in Europe. Today, it's just 26%. And he outlines what needs to happen to stop the European pharma sector from being 'yesterday's news'.

[Watch video](#)

Author

Diederik Stadig

Senior Economist, Healthcare & Technology

diederik.stadig@ing.com

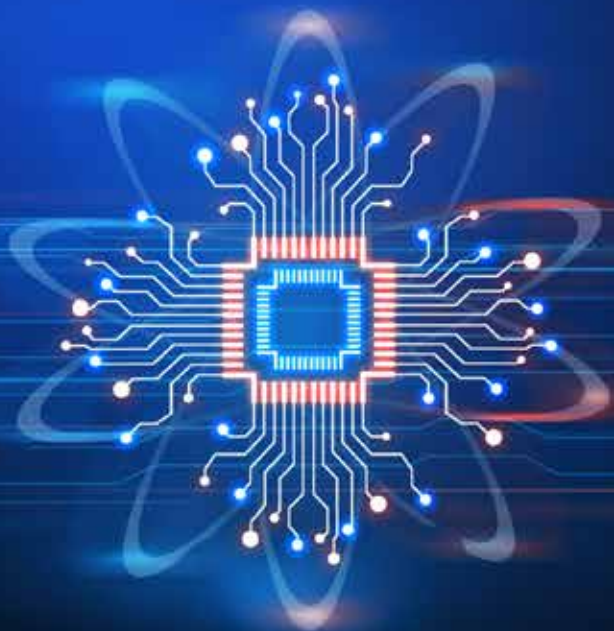
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MANUFACTURING INTELLIGENCE

Exploring the Spectrum of AI Use Cases

JUNE 2024



Inside

- 3** The Increasing Appetite for AI in Manufacturing
- 6** AI for Topline Growth and Competitive Advantage
- 8** 75+ Early Use Cases for Advanced AI in Manufacturing
- 9** Current AI Use Cases in Manufacturing
 - 10** Supply Chain Management Use Cases
 - 11** Design and Engineering Use Cases
 - 14** Production, Maintenance, and Safety Use Cases
 - 18** Quality Use Cases
- 20** Information Systems and Cybersecurity Use Cases
- 22** Warehousing and Inventory Use Cases
- 24** Aftermarket and Customer Services Use Cases
- 25** Getting Started and Creating an AI Roadmap
 - 27** AI Roadmap for Manufacturers
- 28** Conclusion
- 29** About the Research
- 30** About Our Supporters

The Increasing Appetite for AI in Manufacturing

Advances in artificial intelligence (AI) over the last 18 months have put AI on a new trajectory for manufacturing. Of course, the sector is no stranger to AI. Manufacturers have used machine learning for decades to teach equipment to perform tasks, and they are increasingly deploying deep learning to take on more complexity. Significant advancements in generative AI in 2022 opened an entirely new set of opportunities for using AI in the industrial domain. Generative AI is about far more than interacting with chatbots and rewriting essays. Building on deep learning architectures, generative AI can complement a manufacturer's machine learning and deep learning models to add even more intelligence to the entire value stream of manufacturing. All of the advances in AI over the last decade (computer vision, speech recognition, generative AI) combine

with other innovations in distributed computing, processing speed, edge computing, and open-source algorithms, to significantly advance manufacturers' journey toward Industry 4.0.

Most companies are early in the process of figuring out where and how to bring cutting-edge AI into their day-to-day operations. As Paul Bouvy, Senior Director of Supply Chain at **Owens Corning**, put it, "We have used AI every day for a long time to control our manufacturing processes. Here, I'm speaking about things like multivariable modeling and problem solving. But nowadays, everyone thinks of AI as just being about generative AI, large language models, and things like that. We're just starting work in that area."

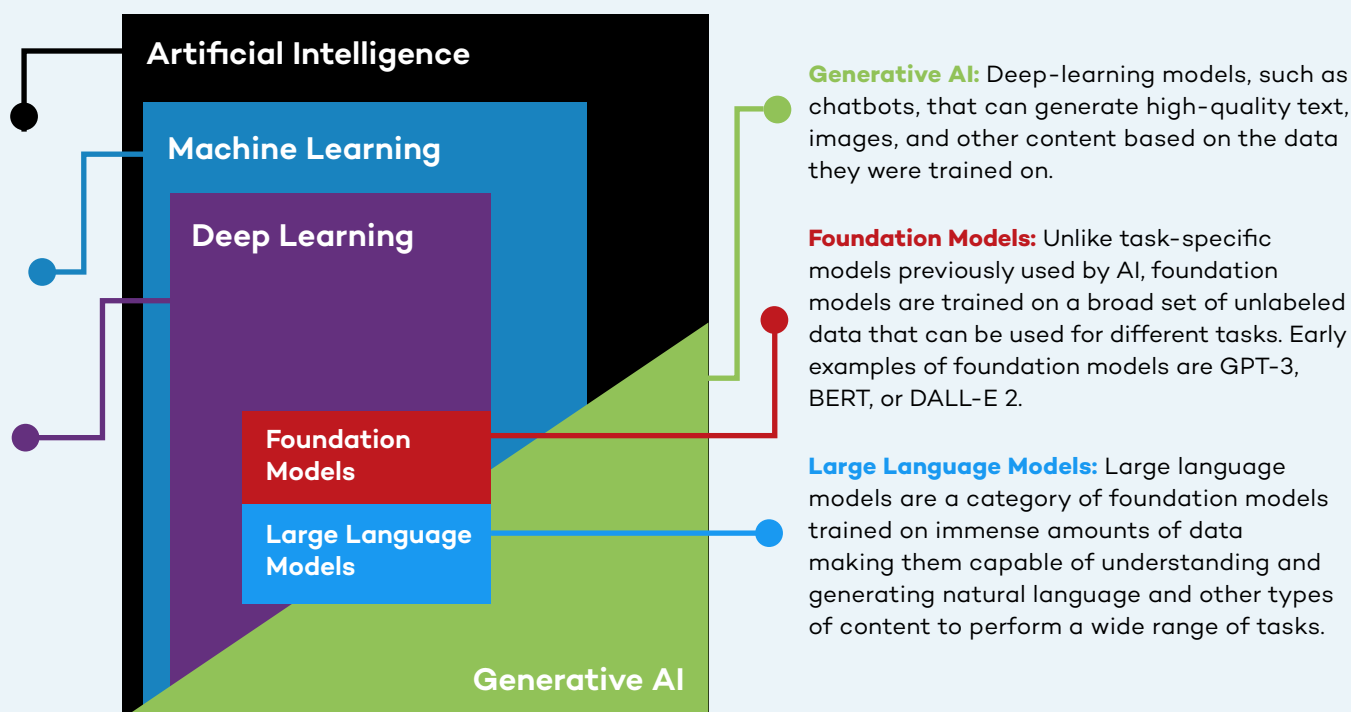
AI Definitions

Artificial Intelligence: Technology that enables computers and machines to simulate human intelligence and problem-solving capabilities.

Machine Learning: A form of AI that enables a system to learn from data rather than through explicit programming.

Deep Learning: A subset of machine learning that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.

Source: IBM: What's Next in AI?

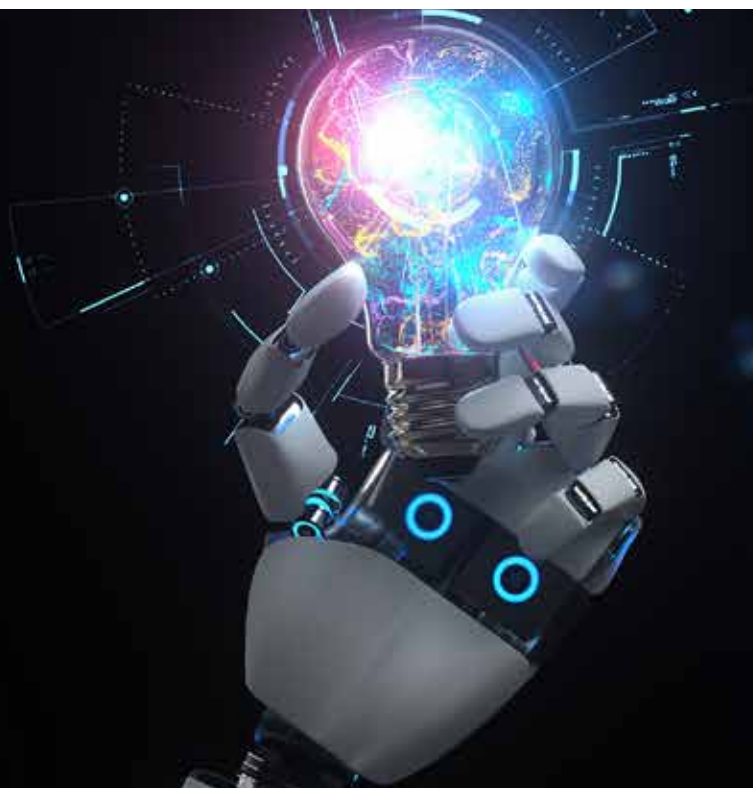


What are the most important new use cases for AI in manufacturing and how quickly are manufacturers deploying them? Manufacturers Alliance Foundation took a deep dive into these questions to better understand the state of play for AI in manufacturing. We spoke with dozens of leaders across a range of manufacturing functions and conducted a survey of over 200 U.S.-based mid-cap to large-cap manufacturing companies.

KEY FINDINGS reveal a strong appetite for making progress with AI among manufacturing companies across dozens of new use cases already operational today. The vast majority (93%) have added new AI initiatives over the last 12 months, and another 6% plan to launch such initiatives soon. Significantly, many AI initiatives are geared toward strategic business gains, not simply cost or headcount reductions. Many manufacturers expect AI to deliver fairly rapid bottom-line improvements in productivity, throughput, and quality.

“We have just scratched the surface of advanced AI. We’re learning how to envision capabilities. Otherwise, we’re imposing limits on ourselves for no reason. If you can envision it with AI, you can achieve it.”

– Katrina Redmond, Executive Vice President and Chief Information Officer, Eaton

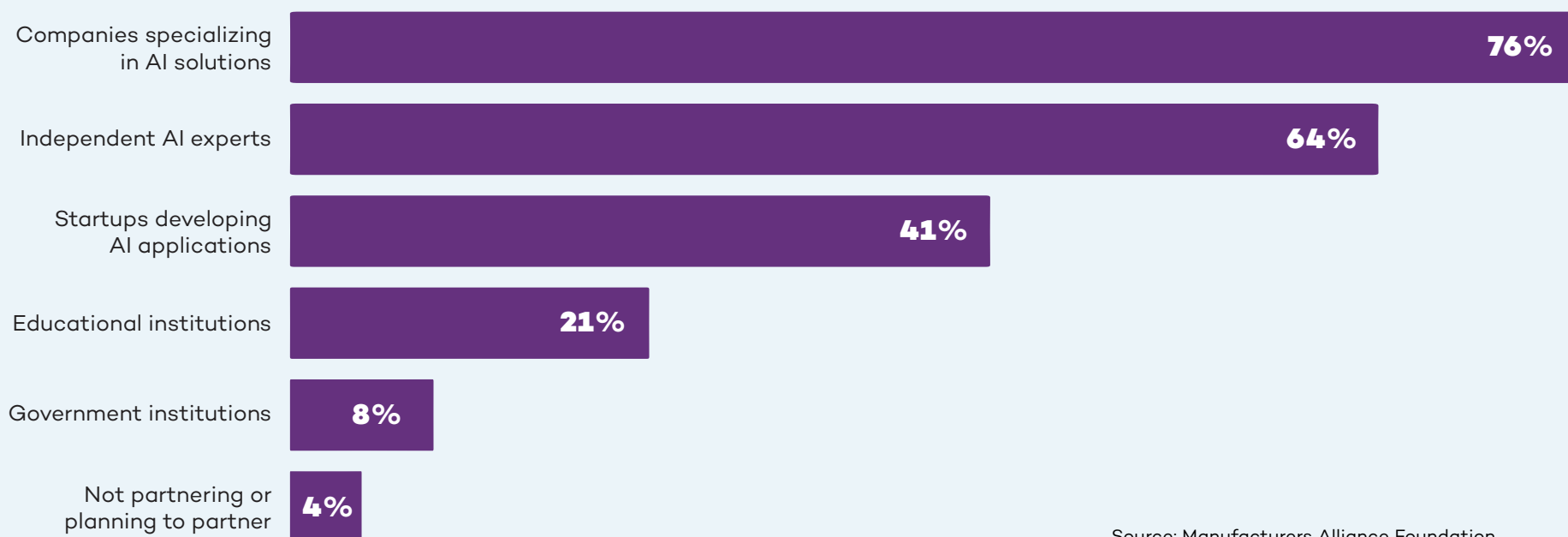


The vast majority (93%) have added new AI initiatives over the last 12 months, and another 6% plan to launch such initiatives soon.

The topline benefits, while expected to take more time, are the holy grail. That's why manufacturers have unleashed citizen developers (business users with little to no coding or IT experience) to come up with as many use cases as possible. As Wallas Wiggins, Vice President Global Supply Management and Logistics at **John Deere** told us, "One of the most exciting things is our citizen development effort where we essentially say, 'Hey, go figure out AI. Do something creative, even crazy. If it works, let's scale it across the entire business. We're trying to drill as many holes as possible because one of these days, one of them is going to pop oil."

The AI learning curve is steep for almost every organization. "I think one of the biggest near-term impediments is that you just don't think about using AI, even if you have it," John Bartho, Chief Information Officer at **Hyster-Yale**, told us. Most manufacturers are looking for guidance for the best use cases but more importantly, how to think strategically about AI and fit it into their technology roadmap. As Joanna Cooper, General Manager of the Mount Holly plant for **Daimler Truck North America** put it, "We can't make this technology transition with the attitude that we will do everything on our own. Daimler Truck has a vast number of partnerships with OEMs and other companies, including AI partners. We have to make progress together."

Where Manufacturers Are Partnering



Source: Manufacturers Alliance Foundation



AI for Topline Growth and Competitive Advantage

Finding new and more sophisticated use cases in the age of generative AI creates a new set of opportunities for manufacturers.

McKinsey estimates that generative AI can contribute between \$650 billion and \$1.1 trillion in annual productivity improvements for manufacturing and supply chain-related activities globally. With this much to gain, manufacturers are playing the long game when it comes to embracing new applications for AI. When we asked how they are prioritizing their AI initiatives, 55% said they are aligning these initiatives to strategic business objectives. Less than a quarter (24%) are focused on potential cost savings.

Manufacturers see cost as less of an obstacle when topline growth and competitive advantage are at stake. As Tim Speicher, Senior Manager of Advanced Analytics and AI at **MSA Safety**, told us: “The initial value is going to be in productivity, such as saving time. Then there are revenue driving opportunities where AI can help us – either in products or in services – and that’s where the excitement comes in.” When we asked manufacturers to rank how they plan to measure the ROI of

Top 6 Expectations for ROI

- | | |
|----|---|
| #1 | Revenue growth |
| #2 | Improved uptime |
| #3 | Product quality |
| #4 | Customer satisfaction |
| #5 | Long-term strategic impact |
| #6 | Performance relative to industry peers, competitors |

Source: Manufacturers Alliance Foundation

their AI implementations, revenue came in as the clear first choice. Long-term strategic impact and competitive advantage against peers ranked fifth and sixth. Cost savings trailed in ninth place.

Manufacturers are in it for the long haul. Very few (16%) expect a quick ROI in under 12 months. For the vast majority, the ROI time horizon is farther out: 42% expect to see it within one to two years, and 40% expect two to four years.

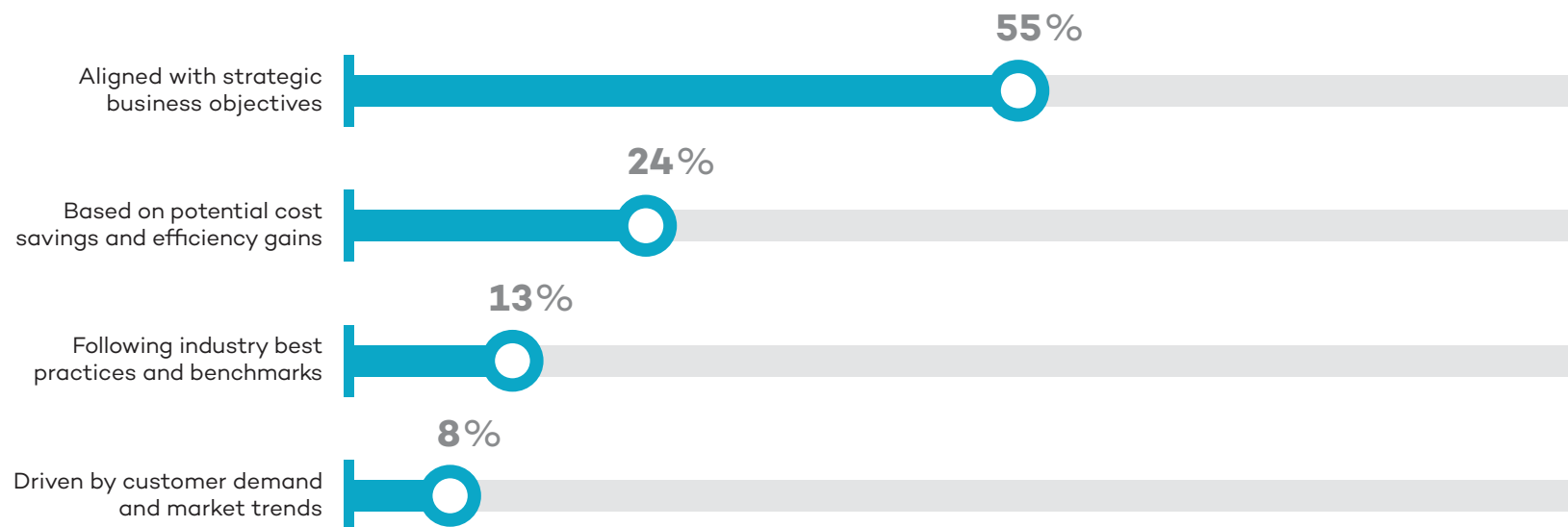
Given the magnitude of change, it will take time for organizations to absorb the impact from AI and realize the full benefits. Karen Powers, Digital Product Transformation Leader at John Deere, talked about just this aspect: “I think probably the biggest challenge in general is not determining where we want to use it, but how we build comfort with it. When will people be able to trust the results and allow the

system to take action for them? The change management aspect of all of this is huge.” Bringing people along will take some time. “We do have executive buy-in. Change is always difficult, so for the business to be totally behind it, we need some quick wins,” Tim Speicher of MSA Safety, told us.

There is a clear recognition that the latest advances in AI are real and here to stay. It will become table stakes just like previous technologies, such as e-mail, Wi-Fi, and texting. As Ben Grimes, U.S. Enterprise Commercial Vice President for Manufacturing at **Microsoft** predicts: “Once someone has it, they won’t want to give it up.” The record-breaking pace of consumer-grade adoption suggests this prediction may prove accurate. Already 27% of American adults say they use AI daily or several times per week, according to **research by Pew**.

AI Investments Focus on Strategic Gains

How do manufacturers prioritize AI implementation initiatives?



Source: Manufacturers Alliance Foundation

75+ Early Use Cases for Advanced AI in Manufacturing

Supply Chain Management	Design & Engineering	Production	Maintenance	Safety	Quality	Info Systems & Cybersecurity	Warehousing & Inventory	Aftermarket / Customer Services
Alternate component optimization	Buyer behavior prediction	Cobots	Anomaly detection	Employee fatigue and distraction	Assembly quality inspection	Attack detection systems	Automated picking	Aftermarket predictive maintenance
Faulty supply tracking	Controller engineering	Energy demand forecasting	Asset health monitoring	Employee PPE compliance	EHS standards library	Attack interruption systems	Autonomous vehicle delivery	Call center knowledge support
Inbound delay forecasting	Design coding assistance	Energy optimization	Contextual predictive maintenance	Potential safety risk prediction	Process parameter optimization	Automated IT operations	Demand forecasting	Connected product diagnostics
Intelligent supplier selection	Design for manufacturability	Machine controller project optimization	Maintenance planning	Real-time monitoring shop floor	Quality prediction	Cybersecurity automation	Dynamic container scheduling	Customer service chatbot
Raw material cost estimation	Design for quality	Material handling optimization	Predictive maintenance	Remote safety inspections	Quality standards library	Data cleansing	Dynamic load matching	Virtual assistance for tech support
Raw material optimization	Design for regulatory compliance	Production line balancing	Root cause analysis	Safety incident analytics	Regulatory compliance inspection	ERP system integration	Dynamic slotting	Warranty claims optimization
Supplier price comparison	Generative machinery design	Production line coding		Safety trend analytics	Visual inspection – cosmetic	Infrastructure performance tracking	Inventory optimization	
Supplier relationship management	Generative product design	Process mining for digital twin setup			Visual inspection – pass/fail	IT service issue prediction	Order processing chatbot	
	Instrumentation data capture	Production planning optimization				Master data management	Predictive stock replenishment	
	R&D efficiency	Robotic process automation				Risk detection optimization	Warehouse automation	
	Verification testing	Spare part inventory				Response automation playbook	Warehouse visibility	
		Virtual expert assistance						

Current AI Use Cases in Manufacturing

There are already hundreds of AI use cases relevant to manufacturing and many of them have passed the proof-of-concept phase to start delivering value. In the past year, **Siemens** has identified 300 generative AI use cases across all manufacturing domains. Del Costy, President and Managing Director, Siemens Digital Industries, US, told us about the company's cross-functional generative AI program which aims to identify efficiency gains, novel product features, and new offerings. "More than 70 of these are already under implementation in a Proof of Value," Costy shared.

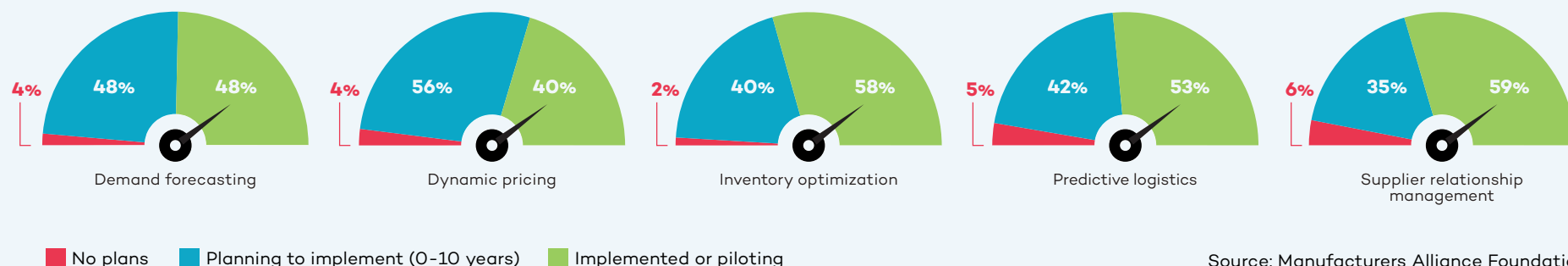
Manufacturers are moving quickly to add AI throughout the manufacturing value stream in areas such as:

- >> **Supply Chain Management**
- >> **Design and Engineering**
- >> **Production, Maintenance, and Safety**
- >> **Quality**
- >> **Information Systems and Cybersecurity**
- >> **Warehousing and Inventory**
- >> **Aftermarket and Customer Services**

This report focuses on seven areas, representing different points along the manufacturing value stream, where AI will grow in importance to manufacturing operations.



Supply Chain Management Use Cases



When it comes to trying new use cases for AI, most manufacturers view the supply chain as a good place to start. AI helps them collect data from across the supply chain in real time. Given the relatively recent disruptions in global supply chains, it is not surprising that supply chain management ranked as the number one functional area for AI initiatives currently among manufacturers we surveyed. “So much of what supply chain management does involves generating content, conversation, validation, and opportunity identification. So, machine learning and generative AI have a ton of opportunity here,” Karen Powers of John Deere told us.

Intelligent supplier selection is already being used to help evaluate a supplier’s pricing as well as resiliency based on current market data. Another application is **supplier relationship management**. As Paul Guerrier, Advanced Technology Center Manager at **Moog**, told us, “I can see us using generative AI tools to evaluate incoming information from vendors to check that the vendor data pack is complete, contains

good performance measurements, etc.” Tim Speicher of MSA Safety envisions AI helping procurement teams “negotiate better contracts and terms with our vendors.”

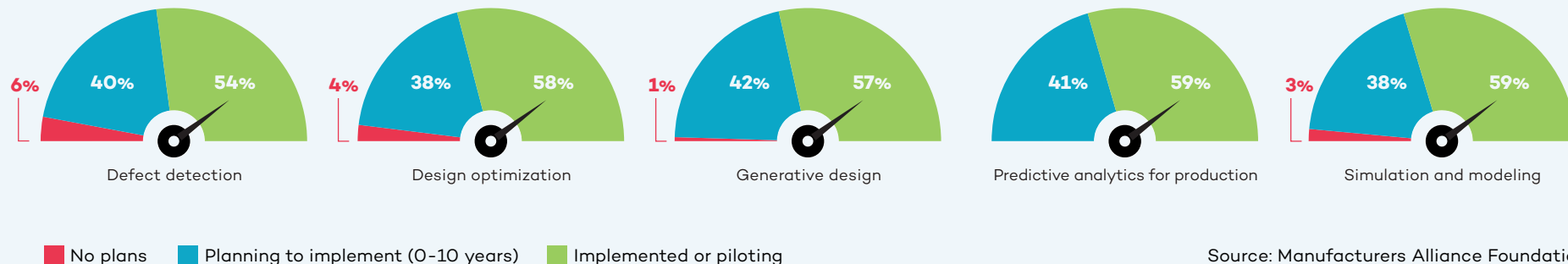
When there is a sudden global shortage of a specific material or subcomponent, AI can be used for **alternate material optimization**. Katrina Redmond, Executive Vice President and Chief Information Officer at **Eaton**, talked about partnering with **Palantir Technologies** to use generative AI for finding alternatives. “All of our information from our ERPs is processed by these large language models. A part shortage or a raw material shortage may be preventing us from completing the manufacture of a product. Do we have alternatives available? By having the descriptions in the system, you can sometimes match on description. If you need an 8-foot rod but only a 10-foot rod is available, we just have to cut 2 feet off to satisfy the requirement for the order.” Marc Rosen, Strategic Accounts, Palantir Technologies, added: “Working with Eaton, we created this AI agent that helps cross-reference parts so that if you have a

\$1 million sale being blocked by \$50 worth of raw materials, it helps them find what else they can use in their inventory to complete the order.”

AI for **raw material optimization** is helping steel manufacturers determine how to use their own scrap in the most efficient way. “As a steel manufacturer, we always have a mix of scrap material on hand as well as the possibility of what that scrap could turn into. So, we’re using AI to help us figure out the probability of producing the product that we want out of this mix. It will help us streamline both the ordering process on the raw material end and the manufacturing process on the melt end,” Jared Noble, Director of Digital Technology for **Charter Manufacturing**, told us.

Looking ahead, one manufacturer is hoping that advanced machine learning and AI will help them predict outcomes of a range of end-to-end supply chain decisions to help them land on the right choices based on the metrics they prioritize.

Design and Engineering Use Cases



For design and engineering, the current use cases are designed to speed up the innovation process and reduce time to market through things like AI assistance in **product design, innovation design, machine design, and regulatory compliance**. Michelle Drew Rodriguez, Partner at **Roland Berger**, discusses innovation efficiency in her work consulting manufacturing clients: “We have already identified more than 200 AI use cases for manufacturers, including in supply chain, procurement, operations, human resources, and more. For example, in the product development and engineering space, we have been working extensively

More than three quarters of manufacturers are starting to use AI for regulatory compliance.

Source: Manufacturers Alliance Foundation

with clients on optimizing product design as well as the product development process. We’re using AI to review the entire R&D portfolio, then systematically applying AI to advance R&D efficiency throughput and deliverables, including reducing the time to develop commercially viable products. This has enabled us to create efficiencies such as reduction in R&D spend (25%+) as well as greatly improved throughput (up to 60% reduction).”

One large manufacturer is using AI to look at **design for manufacturability** of a product to ensure that any errors in design are caught before the manufacturing of a new product or product version begins. Design for manufacturability is especially important in the consumer electronics manufacturing space, where miniaturization must be balanced against the angles and spaces that are required for assembly, especially if that assembly is being handled by a robot.

More than half (57%) of manufacturers are starting to use AI with generative design

and another 22% are planning to start in the next two years. Significantly, manufacturers with more than \$10 billion in annual revenue are much more likely to be focused on AI for generative design, R&D, and quality than manufacturers in lower revenue ranges.

Southco sees potential future applications of AI during the **advanced product quality planning (APQP)** phase. Deep learning AI algorithms can assist with designing for quality by helping the engineer select the right materials. Shailesh Patel, Director of Quality at Southco, described the situation: “If an engineer tries to design a product in a certain material, AI tells them ‘Don’t use that material.’ This is based on past data. We have had a ton of issues with some specific problem materials, so this is very important for us.”

Design for regulatory compliance is another promising application for AI. Southco has started exploring the potential for AI in this area. As Jim Ford, Vice President of Engineering at Southco, explained: “We have to sift through a lot of compliance

“We don’t have to spend hours writing the code. From a product development perspective, it’s huge. We use it every day.”

**– Tom Hower, Vice President of R&D,
Heavy Duty Transportation Group,
Marmon Holdings**

paperwork, and it is a very manual process with people parsing that data. So, we want to scrape the relevant information out of everything that is coming in so they can manage the compliance elements.” John Deere is also using AI to assist with regulatory compliance by providing AI-driven prompting systems that interrogate the regulatory documents to assist with the interpretation of standards.

Southco is just starting to look at the potential uses for AI in **generative product design**. “A lot of our products are very modular. So, we’d love a customer to come in and say, ‘Hey, I like this product, but increase that grip by 10 millimeters.’ And if we could do some of that through AI, potential savings and customer responsiveness for us would be pretty significant,” Jim Ford of Southco told us.



AI in Verification Testing

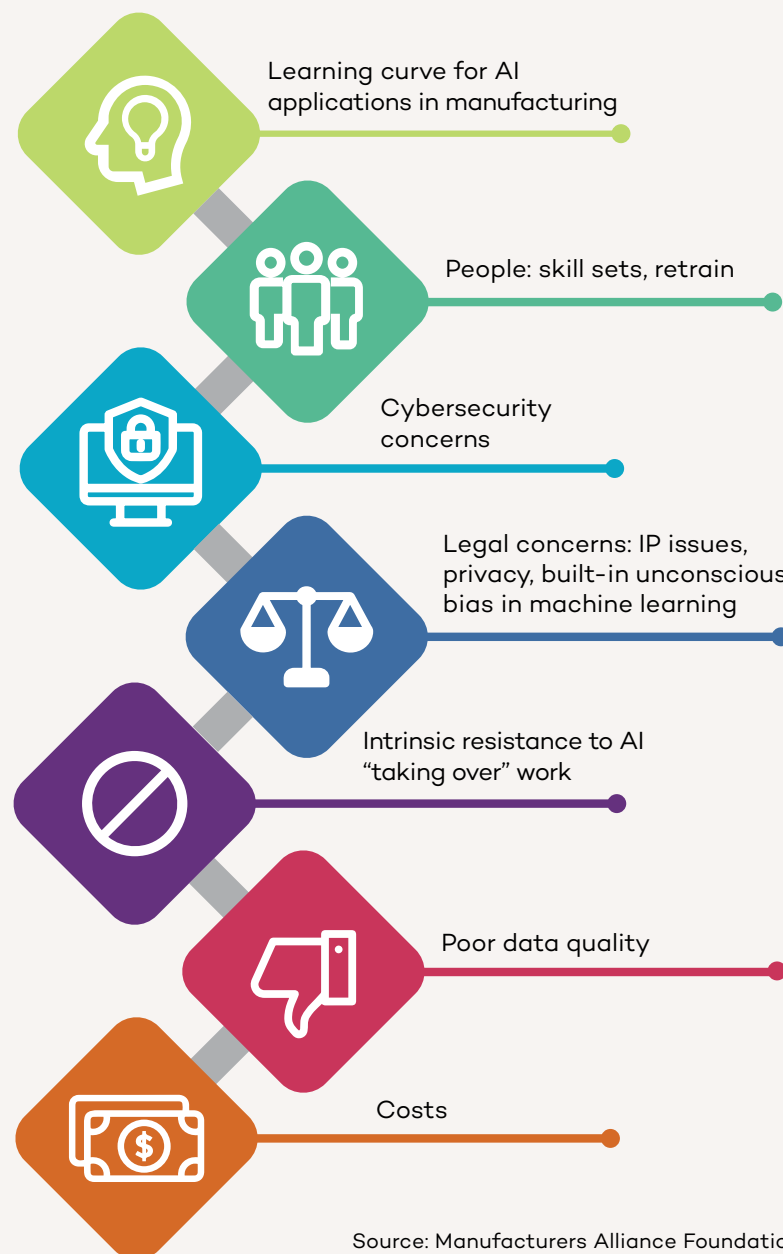
Verification testing is fertile ground for AI. More and more products have connectivity to other products, machines, or cloud applications, and this requires much more involved testing. The key is to develop efficient testing models. As Tom Salapow, Executive Director, Research & Engineering at MSA Safety, put it: “When you have devices connecting to the cloud and to each other, you have a bigger ecosystem and the instances of things you have to verify goes up exponentially. Instead of doing 10 tests, we now have to do 1,000 tests. So, we’re looking at how to develop design of experiments to narrow that down, so instead of doing 1,000 tests, maybe we can do 100 effective tests and cover everything. I think that’s where AI can help us.” Salapow pointed to the magnitude and speed of change in this particular area: “It’s something we’re evolving into now. We have built a team of R&D verification engineers. That wasn’t even a role five years ago. That’s one of the challenges they’re working on now. Any time you make a change, it has a significant effect across everything. You almost have to keep this thing running in real time as things are changing. And that’s a natural fit for AI.”

Several manufacturers talked about using generative AI for **product development coding**. As Tom Hewer, Vice President of R&D, Heavy Duty Transportation Group at **Marmion Holdings** explained: “One of the places we’re using AI is to help our engineers with coding. My mechanical guys are a lot better at software now that they’re using AI because they can say, I want a Python script that does this or that, and the AI generates a baseline script immediately that just needs to be adjusted. We don’t have to spend hours writing the code. From a product development perspective, it’s huge. We use it every day.” Ben Grimes of Microsoft has seen more than a 40% increase in code development among its customers using its GitHub Copilot, which helps developers generate code more quickly.

Looking toward the future, manufacturers are also thinking about AI not only for products but for **generative machinery design**. One manufacturer shared: “Can we use AI to help us design not just our products, but also the machines that make our products? Is AI going to come up with things that a human sitting down with a piece of paper is not going to consider, such as transformative, non-standard configurations of parts and machines?”

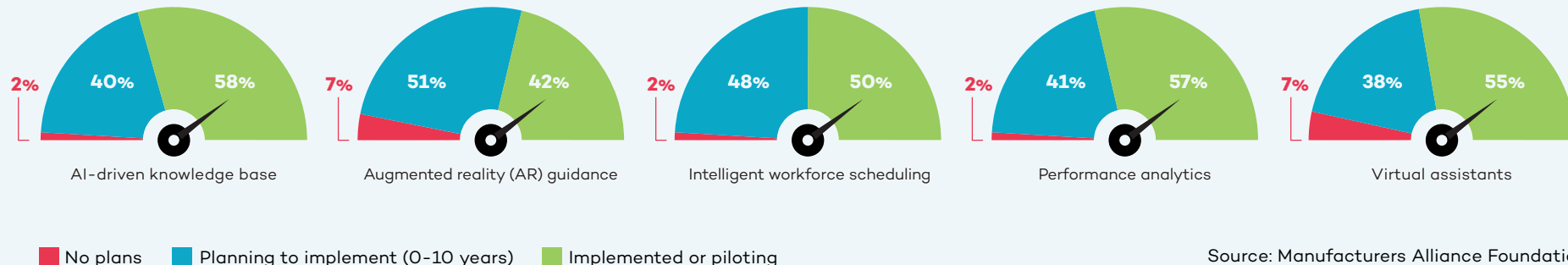


How Manufacturers Rank 7 Key Obstacles to Adopting AI



Source: Manufacturers Alliance Foundation

Production, Maintenance, and Safety Use Cases



The range of operations in the production and maintenance space opens up a wide variety of new AI use cases. Most manufacturers we spoke with have started using generative AI to provide **virtual expert assistance** to production and maintenance teams. Our survey also revealed that 55% of manufacturers are starting to use virtual expert assistants in this space currently and another 16% are planning to do so within the next two years.

Generative AI also has a role to play by providing detailed instructions for maintenance teams including lists of spare parts and images. This not only speeds up remediation but enables less experienced staff to address maintenance problems.

Clarios feeds a large quantity of engineering documents in different formats (PDF, PowerPoint, Excel) into an in-house generative AI system. Sami Khan, Senior Manager for IT Platform Services at Clarios, told us: “We wanted the AI system to ingest about 1,000 different documents rather than having people searching through hundreds

of thousands of pages to get an answer. The chatbot spits out a short answer based on the specific question that is asked. We also want this data to be refreshed periodically as people upload more documents so that it is never out-of-date.” Khan provided a specific example: “You can ask the system ‘what is the procedure for testing paste density?’ The system generates a short answer based on the documents that were ingested and links to the source of the answer. If you want to dive in further, you can click on the source.”

Krishnan Alampallam, Vice President of IT Applications at Clarios added: “We have the same equipment in multiple different versions all over the world, but we also have a constant churn of people. We never want to depend on an individual brain to deliver these answers because then it becomes tribal knowledge. How do we get rid of tribal knowledge? That’s the whole goal behind it.”

Coding assistance is another early application area for manufacturers. This is especially relevant for operations employees who have decades of hands-on experience

but little training in coding. Siemens and its motion technology customer **Schaeffler** are partnering to use generative AI to help Schaeffler’s **automation engineers generate code** for programmable logic controllers more efficiently. This enables engineering teams to generate code faster, with less effort, and with fewer errors. “In the past, we had to speak to machines in their language. With the Siemens Industrial Copilot, we can speak to machines in our language. In a few years, AI will be omnipresent in industry,” Siemens Digital Industries global CEO Cedrik Neike predicted.

AI-enabled predictive maintenance is raising the bar on how assets can be protected in an industrial environment. A great example is AI-based **asset health monitoring** for industrial robots. AI can monitor robot health as well as predict failures that might happen before a scheduled maintenance.

BMW is using AI in the assembly process to analyze data generated by its conveyor technology system. Without any additional sensors or hardware, the company transmits data to its own predictive maintenance

cloud platform where algorithms scan for irregularities that could bring production to a standstill, such as changes in power consumption, abnormal conveyor movements, or illegible barcodes. With AI, BMW is able to identify and avoid faults that would have caused 500 minutes of disruption per year at one assembly plant. With one vehicle rolling off the assembly line every 57 seconds, each minute counts. In addition, BMW is able to use the system's findings to continuously refine and improve the algorithms. The vision is for this system to learn to estimate the time between detection of a fault and potential stoppage, aiding operations teams in determining how to prioritize maintenance tasks.

When these types of artificial intelligence and machine learning are combined with generative AI, manufacturers are able to create a **conversational predictive maintenance** user interface facilitating a natural language interaction between the AI platform and maintenance experts. This dialogue streamlines the decision-making process and saves both time and money.

The synergy between AI and human capital is valuable. Del Costy of Siemens stressed the importance of integrating "human expertise and machine learning to extract relevant information from the equipment and process status." By analyzing the process around the equipment, it is possible to detect equipment behavior changes early, enabling better operating and maintenance decisions.

Given the power of AI in this space, it is not surprising that half of manufacturers are

starting to use AI for predictive maintenance analytics and another 25% expect to do so within the next two years. **USG** is using AI for "**anomaly detection** of both our assets and our processes," Lou Stocco, Senior Technology Manager at USG told us. "It identifies trends that are less than desirable. It could be an issue with mechanical, electrical, maintenance, or production. We also look at raw material consumption, energy usage, and all of the things associated with the overall manufacturing process. We're calling it condition-based manufacturing," Stocco said.

Advanced Technology Services (ATS) is already looking at what AI will be able to do in terms of **contextual predictive maintenance** as capabilities evolve. Micah Statler, Director of Industrial Technologies at ATS, talked about the next steps: "We're looking to go deeper into the predictive and the condition-based realm and feed our models more contextual data. It's one thing to say the machine is vibrating with this signature. It's a whole different thing to say it's happening under these conditions." Statler's vision is "to drive engineering change without the failure of the equipment to begin with. This prolongs the life of the asset."

In the industrial engineering phase, AI tools can help manufacturers and machine builders optimize their designed operating efficiency, desired throughput, and machine availability by running AI-based optimization tools on the simulated machine controllers (hardware in the loop simulation) in high-frequency. The German company **plus10** provides **machine controller project optimization** software helping companies shorten ramp-up

phases significantly by deriving throughput potentials on a controller step-chain level at a very early phase. This reduces later-on mean time to repair (MTTR), production variability, and overall machine ramp-up time of complex production or assembly lines. CEO Felix Georg Mueller told us, "Our software is learning live on machine controllers, either simulated ones during design phase or later on real-ones integrated in the line. By learning behavior models based on high-frequency data streamed in milliseconds, we formalize the behavior, reason the models, and provide counter actions in case of stops, disturbances, short stops, or scrap parts. That generates plus 10% or more output on average, hence our name." When one machine is behaving differently than another in terms of output or frequency of failure, an additional optimizer product can determine the root cause. "We can benchmark machines with each other even if they are thousands of kilometers apart to learn behavior and optimize on the go. In the past, this had to be done manually by engineering, improvement teams, or system integrators," Mueller said.

Manufacturers are also evolving from using AI with individual machines or lines to large-scale **production planning optimization**. Eaton's Katrina Redmond talked about the challenges and the solution. "We have 30,000+ different products. We always need to know how, when, and in what bundles they are selling. AI has given us better clarity across all of our different ERP environments to know which items we are clear to build. For example, it tells us if we have all the product available in the quantities necessary to fill order X, Y, or Z. So, AI is really helping

the planners and the production floor teams figure out which products they can complete before getting to step one of the manufacturing or assembly process. Essentially, AI is telling us, 'Don't bother with this order yet because you're going to be missing a piece at step four.' That allows us to go to the next item where we have all parts and material available."

Material handling optimization is another promising area for AI because it brings together a wide range of data points such as geography, mobility, wireless communication, supply chain, and more. More than half (57%) of manufacturers we surveyed are starting to use AI for material handling and 22% have it on their radar to implement within the next two years. Hyster-Yale is using AI for kitting. John Bartho described what are essentially "shopping cart kitting areas where people are building kits for assemblies which are scheduled to come into production next." The impact is significant in terms of throughput, quality, and cost. "We identify parts shortages earlier and maximize the efficiency of the assembly line. The quality improves because the less you interrupt the process, the higher the quality," Bartho added. This is an entirely new operating model for Hyster-Yale: "Increasingly, our suppliers are either onsite or across the street. This has a financial implication because we're not paying until consumption. The historical model is to buy stuff, put it on the shelf (where it becomes working capital), and then consume it. By using AI for process modeling and relying on more nearby materials, we're more productive and have more working capital efficiency," Bartho said.

Safety plays a role in every function, but it is particularly critical in the production and maintenance domain. AI-based visual systems are already on the market that address everything from **personal protective equipment compliance** to employee fatigue. **Stroma**, for instance, provides edge cameras and industrial edge mini-computers for advanced vision-sensing capabilities to ensure operational staff are wearing the appropriate PPE. Stroma's customers in the energy sector, for example, monitor the use of face shields, gloves, and footwear in real time. "We are tracking safety procedures to ensure they are properly followed in the field. In addition, our system can warn field operators in real time and send those same warning messages to managers," Anil Uzengi, Co-Founder and CEO of Stroma, explained. Stroma's systems can also check for **employee fatigue and distraction**. Uzengi shared an example from a customer in the consumer packaged goods sector: "You can see the worker's attention level is 82%. If it drops below 70%, the system issues a voice alarm to the worker in real time with a customized warning like 'Go grab a coffee!' The worker can respond via the same device to confirm the message was understood. Warnings are captured on a dashboard to measure event trends over time."

One Environmental, Health and Safety (EHS) Director talked about the value of AI-powered **remote safety inspections** in locations where an EHS manager is not physically present on site. "We run extremely lean as a company and we're a smaller manufacturer in terms of operations, so we just don't have the resources to have EHS at every single plant.

AI can help us supplement that, for example by identifying a hazard in an aisle, a blocked fire extinguisher, or inappropriate PPE. I think it's fantastic." They also talked about using AI for **safety trend analysis**. "With AI you can discern from the data if you have a location issue, a people management issue, or a supervisor who is not performing."

Mitigating potentially catastrophic safety events is a priority for every manufacturing company. At **The Heico Companies**, Vice President of Environment, Health and Safety Dave Roberts is focused on **safety incident analytics** with respect to potentially severe incident or fatality or PSIFs, a commonly used measurement in the EHS space. "We're trying to prevent catastrophic events from happening, and we're using AI to help us identify the critical precursors that lead to those major events. The whole idea of tracking PSIFs is to avoid getting bogged down by chasing minor events and be in a better position to prevent a significant incident," Roberts said.

Future safety applications based on AI are an important research area for MSA Safety. As Tim Speicher pointed out: "PPE is really the last line of defense. The first line of defense is not to let something bad happen to begin with. And in between there's all this space where we could use more of the data that we collect from our connected devices. We may be able to predict or project bad behavior before it starts. I mean, we're not talking about the movie *Minority Report* here, but we can start to identify problem areas, problem tasks, or even problem people. That's where our opportunities are."

AI as a Training Accelerator

Most of the companies we interviewed are struggling to keep up with training requirements. Generative AI is being used to address training challenges including new employee onboarding, cross-training of existing staff, and bridging the knowledge gap between early and late career employees.

As Joanna Cooper of Daimler Truck North America pointed out, “AI is an awesome tool, especially in light of the changing workforce. There aren’t a lot of people that come from a mechanical background, people aren’t working on their cars much anymore, and shop classes don’t exist in many schools right now, although some schools are starting to offer them again. Being able to use AI to help employees understand more quickly what good looks like as quickly as possible really helps.”

Felix Georg Mueller at plus10 talked about the ability of AI to reduce the reliance on tribal knowledge when production line issues occur:

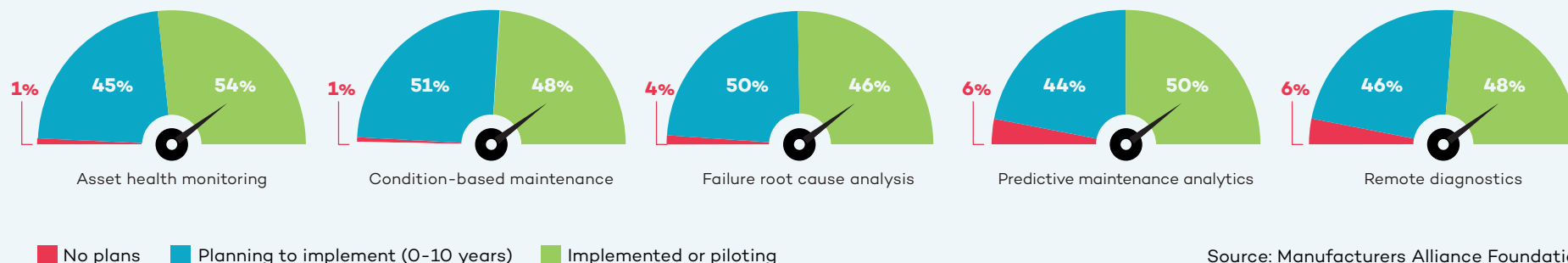
“Let’s say you have an expert who is sitting in front of the line who knows everything by heart. Then you have no problem because he or she can solve the problem in minutes. But those people are quite rare, especially on night and weekend shifts. Employees may not know where to find the root cause or how to address it. There can be a lot of trial and error, and sometimes they make the situation even worse. So our software tells them the root cause related action that needs to be taken and reduces the mean time to repair.” As a result, less training is required because plus10’s optimization tool proactively provides the missing knowledge, which reduces problem-solving time up to 40% in real-world, multi-shift operations.

A leader who manages training for a product design engineering team in the off-highway vehicle industry described the situation at his company. “More than half of our design engineers have been with the company less than five years. Three to four of those years have been spent working from home. So, we’re looking at ways to help these new engineers find the information they need to be successful. In our engineering knowledge world, AI can go out and scour whatever we connect it to and bring that information back. The idea is to teach these R&D engineers how to do their jobs based on our standard processes and tools, but without having to go through years’ worth of development with somebody sitting right next to them.”

In addition to AI-based training, manufacturers are also deploying “train the trainer” campaigns to spread AI knowledge throughout their organizations. Jared Noble at Charter Manufacturing told us about training “30 AI subject matter experts inside functional areas across the company’s business units.” Those people are helping to identify valid use cases for AI including the value it can deliver and the shape of a deployment. “The goal is for these people to have a good understanding of the basics of the tool sets as they’re seeing problems in their space. For us in the data science group, it means we’re getting more valuable information about how to approach use cases. We want to avoid having a divide between the technologists and the people in the function,” Noble stressed.



Quality Use Cases



Source: Manufacturers Alliance Foundation

AI is adding significantly to the technology mix supporting quality. The advances in computer vision over the past decade have already made computers superior to humans in their ability to catch things that are not visible to the human eye. Now, with generative AI, analysis of images can help reduce false positives (good products that fail inspection) as well as false negatives (bad products that pass inspection) thus increasing throughput and quality. More than half (55%) of manufacturers are starting

More than half of manufacturers are starting to use AI for defect detection systems. Another 20% plan to within the next two years.

Source: Manufacturers Alliance Foundation

to use AI for defect detection systems in production. Another 20% plan to within the next two years.

Daimler Truck North America is using AI to ensure **vehicle assembly quality**. “Engineering designs are becoming more complex, so AI enables us to ensure proper torque, orientation, and seating when parts of the truck are being joined,” according to Joanna Cooper of Daimler truck. All of these things need to come together in a timely manner on the line, so it is important that operators understand the desired result. “We’ve taken one of our more complex joints that has the layers of all the things that need to happen correctly to show operators what good looks like. We started with the vision system, and now we’re using machine learning to capture various failure modes to make the vision system even more robust. The idea is to capture those defects as quickly as possible,” Cooper continued.

Chatbots play an important role in quality management as well. “I’m frankly stunned at how good the chatbot technology is. We

John Bartho of Hyster-Yale talked about using AI for **visual inspection and analysis**: “I recently visited our factory in Mexico where we make all of our frames. You might think this wouldn’t be the most automated place, and historically it hasn’t been. There are about 300 employees, 210 of them are welders. But what I saw there surprised me. There are robots doing welding because they can do a seam without interruption thus delivering a higher quality weld. In the quality area, they are using vision systems to do random inspections of frames. The vision system compares the manufactured frame to the CAD drawing and highlights anything which might be variant. This is incredibly helpful because problems are often not visible to the human eye or difficult for the non-trained person to see. As we do more of this, we can improve quality overall by identifying these issues earlier.” As this example makes clear, the AI-assisted visual inspection and analysis is addressing not only quality but also a training bottleneck, an issue encountered by many of the companies we interviewed.



put all of our **quality and EHS standards** and processes into the AI system over a period of a couple of weeks. Now we have an open library of easily accessible information,” Martin Smith, Vice President of Quality and EHS at **Danfoss Power Solutions** told us. He also shared: “People are falling off their chairs at how simple it is to find answers. We tested it in a live meeting by throwing questions at it. It just came straight out with the correct answer. There was silence in the room. This saves literally hours of searching and sometimes only finding irrelevant documents.”

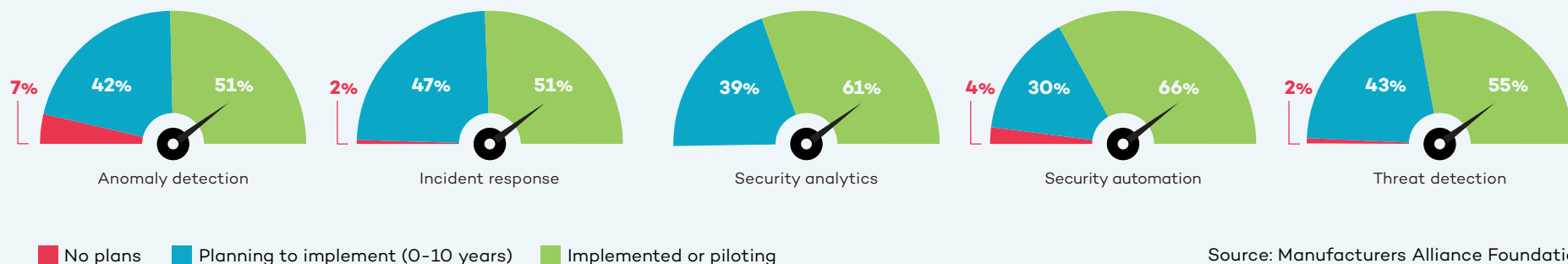
Computer vision for **quality prediction** is another valuable AI use case. Anil Uzengi of Stroma discussed his company’s work for a major tire manufacturer: “We are analyzing rubber sheets and detecting defects in real time. Our advanced vision capabilities use industrial edge cameras, and AI predicts anomalous areas. This triggers a warning to an operator or manager about the need to shut down the machinery.” Uzengi highlighted the incremental but significant ROI impact delivered by AI in this application: “This

company already enjoys 96% to 98% accuracy in their defect detection process, so we are just adding another 0.5% to 2% into this 96% to 98% accuracy. The incremental improvement in quality provided by AI may sound small, but for a large company like this, it means millions of dollars per year.”

AI can also take the subjectivity out of some visual inspection ratings, such as **cosmetic inspections**. For example, a specification for a bright zinc coating can be tricky without AI because humans may disagree about the definition of bright. Similarly, pass/fail cosmetic inspections of machine components can also be a matter of interpretation. Paul Guerrier of Moog shared: “In aerospace many parts are anodized, and pass/fail inspections look at things like dents and scratches.” It is virtually impossible to prevent them completely, but they can be minimized. Guerrier talked about potentially using AI “to define the exact criteria (size of flaws, for example), for pass/fail decisions. I think you can actually define that within the vision system.”

Moog started using AI for quality to increase throughput and to address cross-training requirements for its operators. Paul Guerrier described layering computer vision and deep learning into traditional visual inspection practices: “I would describe it as **AI-enabled inspection**. Our operators have traditionally checked things through a microscope. We added a camera as well as a computer to do the deep learning. We then created the hardware data set and the training images which were used to train the AI. Now the operator can either look down the microscope or they can look at the flat screen, which is displaying the image from the microscope in pretty much real time. If they look at the flat screen, the AI gives them pass/fail advice while they’re inspecting. The human is still totally in charge.” As Guerrier described it, this example of using AI is his “Goldilocks scenario” because it is an augmentation, not a replacement. “If the AI stopped working or started misbehaving, the operator still has control and we can always revert to manual,” Guerrier said. Moog has seen a significant productivity improvement because of this change. In addition, ergonomic safety is enhanced because employees can spend less time bending to look through a microscope.

Information Systems and Cybersecurity Use Cases



Digitalization is increasingly turning manufacturing companies into data-driven enterprises. Security and data quality are paramount and growing more challenging as manufacturers add more sensors, edge devices, autonomous vehicles, and other connected equipment to the factory floor. AI can help manufacturers address this complexity as well as monetize their data.

The key first task for most of the companies we interviewed is addressing the widespread problem of data integrity. **Data cleansing** is job one because data must be accurate, complete, and in context to be useful. As John Bartho of Hyster-Yale put it: “We have a lot of data, but when we first put it all together in the same pot, the stew didn’t taste good. Now we have a centralized group for master data management. We started with all the data around the customer, and we’re expanding that.”

Vitaalic’s Director of AI, Bob Straight, talked about aligning programs to fit an organization’s data strategy maturity: “I work

closely with our data team to make sure that I’m not overcommitting us if the data isn’t ready.” AI is helping companies address their data quality and data strategy weaknesses. AI-driven **master data management** can automate the monumental tasks of profiling, cleansing, and standardization.

“We are only scratching the surface in what generative AI will mean to security operations.”

– Kevin Faulkner, Director of Product Marketing, Fortinet

This is a critical step because the data will be used to train the AI model. If data is inaccurate or mislabeled, the output will suffer. One manufacturer shared, “We

created something like a **customer service chatbot** tool that uses AI to generate answers. Basically, the whole system worked great, but a lot of the answers were wrong because our source data wasn’t in the right place, was inaccurate, or wasn’t up to date. The weak link in our tool was the data that it was pulling from.”

Many companies have grown through acquisition and need to make disparate information systems work together seamlessly. Katrina Redmond shared Eaton’s application of AI to address **ERP system integration**: “We have so many feeder plants with different systems, and the information doesn’t clearly flow together. But with AI you can load all the individual ERPs into a system and then cross-reference everything more simply.”

Getting systems to connect while securing them against cyberattack is essential. Two-thirds of manufacturers are using AI-based **cybersecurity automation** to proactively defend against attacks before there is a breach.



AI and machine learning are well-entrenched in this space and have been “for more than a decade in the form of processing billions if not trillions of data points every day,” Richard Springer, Director of Marketing for OT Solutions at **Fortinet**, told us. “The conversation around generative AI is relatively new, but that’s the future. It’s not going to replace humans in this space, but rather make them more powerful and efficient.”

Fortinet has been adding generative AI assistants to its products since late 2023. Kevin Faulkner, Director of Product Marketing at Fortinet, explained, “this is not about opening a second browser window for Chat GPT and asking questions. This is actually embedded within our products. It is context relevant and context aware, including very

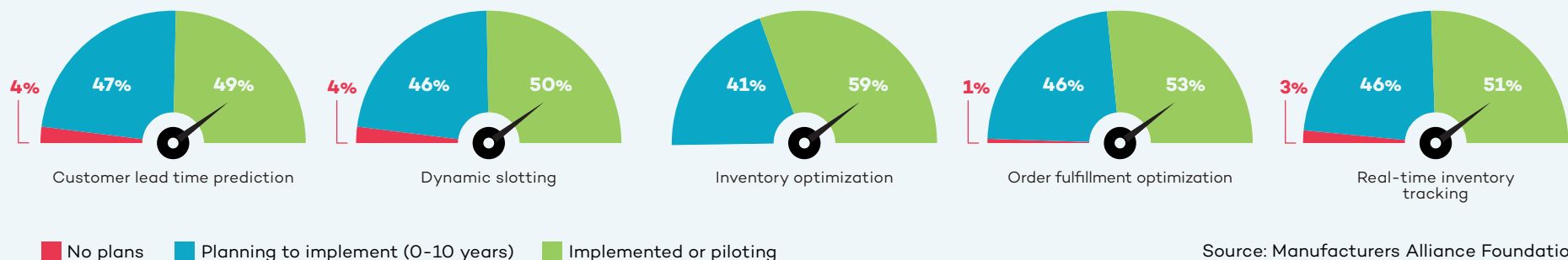
specific OT knowledge.” In the case of an **intrusion detection**, an employee would typically perform **threat analysis** including determining how dangerous it is, what it typically does, and what techniques can be employed to investigate the intrusion. “With our solutions, employees can use either a written or verbal interface to get answers to these questions. This makes people better at their jobs by allowing entry-level analysts to get more work done at a more senior level because they have an assistant looking over their shoulder,” Faulkner continued.

Large language model integration also makes it possible to generate recommendations on how to prepare and respond. As Rod Locke, Director of Product Management at Fortinet, told us, “You can actually build **automation playbooks** based

on best practices for a given scenario, including processing certain types of events in an automated way in the future.” The result is effectively upskilling employees in terms of their effectiveness to the organization.

Time is critical to security. When bad actors find out about vulnerabilities, they target them very quickly. “There used to be about an eight or nine day time period to prepare after a vulnerability was released. Now, that has been shortened to about four or five days,” Richard Springer told us. “It can be a very manual process to separate the wheat from the chaff and find out what’s actually going on. With generative AI tools, we’re making employees faster and elevating them to a higher-level response allowing us to shrink the time it takes to get in front of bad actors.”

Warehousing and Inventory Use Cases



Advanced AI is enabling manufacturers to solve a number of warehousing, inventory, and logistics challenges in ways that were previously impossible. **Warehouse management automation** systems are changing the way inventory is tracked, picked, and replenished.

A great example is **inventory optimization**. More than half (59%) of manufacturers are starting to use AI for inventory optimization and another 26% plan to within the next two

years. When a manufacturer announces a coming price increase, the result can be a spike in orders followed quickly by a backlog. To avoid this situation, manufacturers are using AI to prevent this type of avoidable inventory volatility. Bob Straight of Victaulic told us that inventory forecasting is a focus area for his team: “Addressing inventory forecasting challenges represents a significant win for Victaulic because it benefits not only operations and supply chain, but also sales and customer care.”

USG considers inventory management a strategic component of its value proposition as a company. **Dynamic slotting** plays a key role. As Lou Stocco explained it: “We are a make-to-stock manufacturer, so our guiding strategy is to be the lowest landed cost producer in all of our product spaces as well as the easiest to do business with. We are using AI to better align the amount we produce with the limited space in our warehouses. The idea is to get as much product to as many customers as possible



“We are transitioning these people from non-value-added work to value-added work which allows them to focus on parts of the operation that really need it.”

– Lou Stocco, Senior Technology Manager, USG



while giving them excellent service. For some of our customers, we guarantee next-day delivery. That level of service requires careful orchestration of a lot of product on the shop floor.” The big picture for USG is to have employees engaged in more valuable activity instead of extracting data, creating dashboards, or doing analysis. Stocco explained: “We are transitioning these people from non-value-added work to value-added work which allows them to focus on parts of the operation that really need it.”

When it comes to logistics, AI is playing new roles as well. John Deere, for example, is looking at AI solutions to become more efficient with its delivery trucks. Using generative AI for **dynamic load matching** may be the answer. As Wallas Wiggins of John Deere explained, “We would like to have a truck out there to deliver and pick up multiple times before it circles back home. I see that as one of the next big fronts for us because we spend a lot of money on logistics. We also have sustainability goals. I want to make sure every time a truck rolls, it’s meaningful rolling, not empty rolling.”

Process mining is playing a key role here. The German software company **Celonis** has developed process mining software that creates a digital twin of a company’s processes by capturing data from every split second of a process and turning it into a digital twin. Their work with **Mars** on dynamic load matching has achieved significant results. Mars has been **able to combine truck loads**, cut shipping costs, and speed up delivery resulting in an 80% reduction in manual touches as well as lower costs and reduced emissions.

Aftermarket and Customer Services Use Cases

Many manufacturers are exploring how to build deeper relationships with customers through service excellence and as-a-service offerings. AI can play an important role by giving manufacturers insights through **connected product diagnostics** and **always-on self-service assistants**. These tools give manufacturers actionable data about their products (how they are performing, how customers are using them) that can be plowed back into R&D and product development.

Traditional technical support is getting faster and better through **virtual expert assistants**. Siemens has deployed technical service chatbots that are able to understand questions and simulate human conversation. Katrina Redmond of Eaton talked about the challenge of technical support for products that are 40 or 50 years old and still in use. “We’re using AI to quickly ingest all of these manuals and decades worth of information so the tech team has it at their fingertips. Because of AI, a technical support person can answer a phone call from a customer and immediately respond with the best answer for their particular issue. Turnaround times can be dramatically reduced.”

One heavy equipment manufacturer is using AI for **warranty claims**. An executive there shared: “There’s a lot of free-form text in the warranty claims, several hundred words in each claim. Individuals used to spend an enormous amount of time reviewing these claims. Now we have deployed a large language model that simplifies the claim

information. We have just rolled this out, but it has the potential to save hundreds if not thousands of hours of people’s time looking through warranty claims per year.”

Danfoss is testing AI to omni-channel customer feedback arriving via email, call centers, and customer portals. “In the past, addressing customer feedback meant manually searching for product number, serial number, manufacturing site, point of sale, etc. I thought deploying AI to handle this would be hard, but it was remarkably easy. We have the same amount of information as we had before and didn’t lose anything,” according to Martin Smith at Danfoss Power Solutions. This allows employees in these roles to be re-deployed into value-adding parts of the organization.

Aftermarket predictive maintenance is another promising area for manufacturers. New entrants such as Tesla and Rivian are offering the ability to predict when a part might fail based on data they capture from the vehicle itself. This information can be enriched based on actual miles driven, vibrations, geography, climate, road conditions, and driver behavior. By correcting problems earlier, customers will likely save on repair costs and automotive companies will encounter fewer warranty claims.

Astec Industries is putting more and more intelligence into its products along what it calls the rock-to-road value chain. Eric Baker, Vice President of Astec Digital, talked about the importance of speeding innovation

and making its products smarter “because customers are demanding it.” One example is inspection of asphalt silos being used by customers involved in building or maintaining roads. “There is a lot of weight in those silos, and the material is abrasive. Silos need to be inspected to ensure that they maintain their structural integrity,” Baker said. With a robot or drone remote control device, Astec’s customers are able to visually inspect silos and process the information using AI. “Previously, analyzing that information had been a manual process, but with machine learning and AI, we are able to identify thin spots or worn-out welds to determine if a silo is structurally sound.” Baker talked about the growing importance of digital twins for this traditionally low-tech space. “We are investing heavily in augmented reality so that it is possible to see one piece of a plant that exists in the physical world and add other pieces in the digital space, allowing customers to visualize how things fit together,” Baker said.

Caterpillar is using aftermarket predictive maintenance to connect dealers with customers’ equipment. Its AI systems are analyzing **data from more than 20 different sources**, such as equipment sensors or fluid analyzers, to create an accurate picture of the performance and health of the customer’s asset. This allows a Caterpillar dealer to monitor the equipment, detect possible issues, and alert the customer to take action before a failure occurs. The dealer can then recommend the parts and services needed to keep the equipment running.

Getting Started and Creating an AI Roadmap

A Conversation with Michelle Drew Rodriguez, Partner, Roland Berger

Q **Manufacturers want to know how to deploy AI as quickly as possible. What do you recommend when you consult with manufacturing clients eager to get started?**

A First, we try to make the process as easy and pragmatic as possible – “less hype, more results” is our motto when it comes to implementing AI with manufacturers. We take them through a systematic framework for developing an AI roadmap, and partner with them to 1) assess their current state, 2) define their future state, and 3) create a detailed execution plan. (See roadmap, page 28.) At kickoff, we ask key questions including: What is the current state of their strategy, value proposition, and technological maturity? How are the people, processes, and culture working together in the organization now, and what does its wider ecosystem look like? We ask that companies be brutally honest in their answers; it is the only way to make progress. At this point, we often hear about problems with legacy systems, data silos, and poor data quality in general. This is very typical of many manufacturing organizations right now.

The current state assessment is a great way to get everyone on the same page and using the same vocabulary, which is an important first step. We see barriers to adoption for the simple reason that people may not have a clear understanding of what AI is, or what it can do for them. There’s so much hype around AI right now, and to many people it just seems big and nebulous. At the same time, we encourage companies to start small and find success quickly. What we mean is, once you’ve identified a few easy, low-hanging use cases within the organization, execute pilot projects to gain momentum. Quick success and real proof points help make AI tangible for employees. This is the quickest way to demystify AI and spark interest among people who don’t have much knowledge or belief in it. Demystifying AI and building a strong, unified quest for change needs to happen

early on. This allows companies to build leadership resolve on a strong foundation, including a commitment to invest and drive accountability through the businesses for the changes ahead.

Q **The current state exercise sounds important but also arduous. Do you find that companies want to stop here and clean up the existing problems, or do they want to plow ahead with AI?**

A From our experience, it’s a mix of both. Companies are finding that AI isn’t so much a new challenge but more of a solution to many of the problems they have identified in their current state analysis. Take the siloed data problem. AI can help manufacturers share data from disparate systems across multiple entities to make more sense of, for example, supply and demand trends. They get better at fulfilling customer needs and optimizing inventory instead of guessing or simply creating more buffer.

When companies start to envision their future state with AI playing a role, they also develop a clearer AI vision aligned with broader business goals. For example, they are not just brainstorming where they can add AI because it is technically possible; rather, they are prioritizing use cases based on business impact. This makes it easier, at this point in the process, for manufacturers to derive technical requirements and blueprints as well as needed organizational and governance structures.

The final step is execution planning. This is where manufacturers have an overall actionable plan as well as methods and KPIs to track the value that AI is delivering. They are making further refinements in terms of technology, people, processes, and culture, and are also documenting early wins and weaving any lessons learned back into the overall plan. Once they start gaining traction, companies can take a hub and spoke approach to saturate the businesses and functions with the knowledge and success already

gained. We find that creating and celebrating the early, quick wins really helps the broader organization's receptivity to implementing additional AI measures.

Q Based on your experience with manufacturers so far, how fast will AI spread in manufacturing?

A Nobody has a crystal ball, but I think it will be very similar to other technological transformations and will evolve at a much faster, exponential pace given the VUCA (Volatile, Uncertain, Complex, Ambiguous) times manufacturers live in now. Simply put, the fundamental need for real-time decision making is greater than ever before, and the stakes perhaps higher than ever. Therefore, I believe the typical barriers to adoption are lower for AI than prior technology adoptions. Use cases that make sense now will continue to make sense, but the solutions keep getting better and more all-encompassing. For example, use cases today are often aimed at solving one problem, but as AI maturity advances, manufacturers will be increasingly using platform-based solutions. They'll address multiple use cases at once rather than singular pain points. The impact will be enormous not only in terms of competitiveness of individual companies but also in the ability of manufacturing as a whole to take on megatrends such as resiliency, resource scarcity, and climate change. Manufacturers I speak with care about these issues and view digitalization and AI as key to addressing these grand challenges.

AI Roadmap for Manufacturers

	Current State Assessment	Future State Identification	Execution Planning
Strategy & Value Proposition	Secure leadership support & commitment Evaluate underlying aspirations from AI integration Understand business needs & expectations Gauge investment commitment to drive AI initiatives	Define and communicate clear AI vision & strategy aligning with broader business goals Establish evaluation and prioritization metrics for AI use cases, including applicability and business impact Assess ROI of AI initiatives	Formulate overarching actionable plan for AI integration (PoC, prioritized waves, etc.) Maintain, evaluate & prioritize the living document of AI use cases Develop methods & KPIs to track value delivered
Tech & Execution	Conduct readiness assessment of existing AI technology & processes > Technical blueprint re: architectures > Data availability, quality & governance > Infrastructure & security > AI tools & processes	Define target technical blueprint based on clear guiding principles Identify potential deficiencies/gaps as well as necessary tech partner(s) Establish data assets required for AI tools Design target infrastructure & forecast compute demand with aimed security level	Develop a roadmap to address the deficiencies within the technical blueprint Establish data acquisition processes Determine protocols to address data security & compliance risks Design a scalable infrastructure strategy to support future growth
People, Processes & Culture	Review as-is operating model Identify roles dependencies Examine present governance & policies Measure workforce capabilities and skills Evaluate established partnerships	Develop target operating model, including business process, roles, etc. Design AI governance structures Determine new skills due to AI integration Establish prioritization criteria for potential partnership	Establish tracking KPIs and milestones Integrate guidelines and principles into AI deployment strategy Design talent attraction & upskilling programs Activate prioritized partnership for implementation

Source: Roland Berger



Conclusion

As our interviews made clear, the race to bring more AI into manufacturing is on. It is not a question of if manufacturers will lean more heavily on AI throughout their value stream but how much and how quickly. The already large number of AI use cases is expanding every day as manufacturers open the aperture on how to think about the power of AI throughout the value chain. Likewise, as AI evolves from individual use cases to bundles of use cases that address sets of requirements, the power of AI will expand even more rapidly.

While there is a learning curve for every company, most organizations discover early on that AI helps them solve existing problems (such as training bottlenecks and supply chain issues) more efficiently while increasing safety, productivity, and quality in their operations.

At first glance, AI may seem like another major transformation to address within organizations that are already struggling with digitalization. But **many of the foundational steps required for successful digital transformation are also required to deploy advanced AI.** A mature data strategy leveraging accurate data will be the first major step for many companies and is a precondition for both digitalization and AI. Relatedly, many manufacturers find they need to take a hard look at their existing processes early on so that they avoid the trap of simply replicating a bad set of processes in the

digital realm. Connectivity is also paramount. Artificial intelligence, like digitalization, requires that machines and systems talk to each other in real time to provide full transparency into operations.

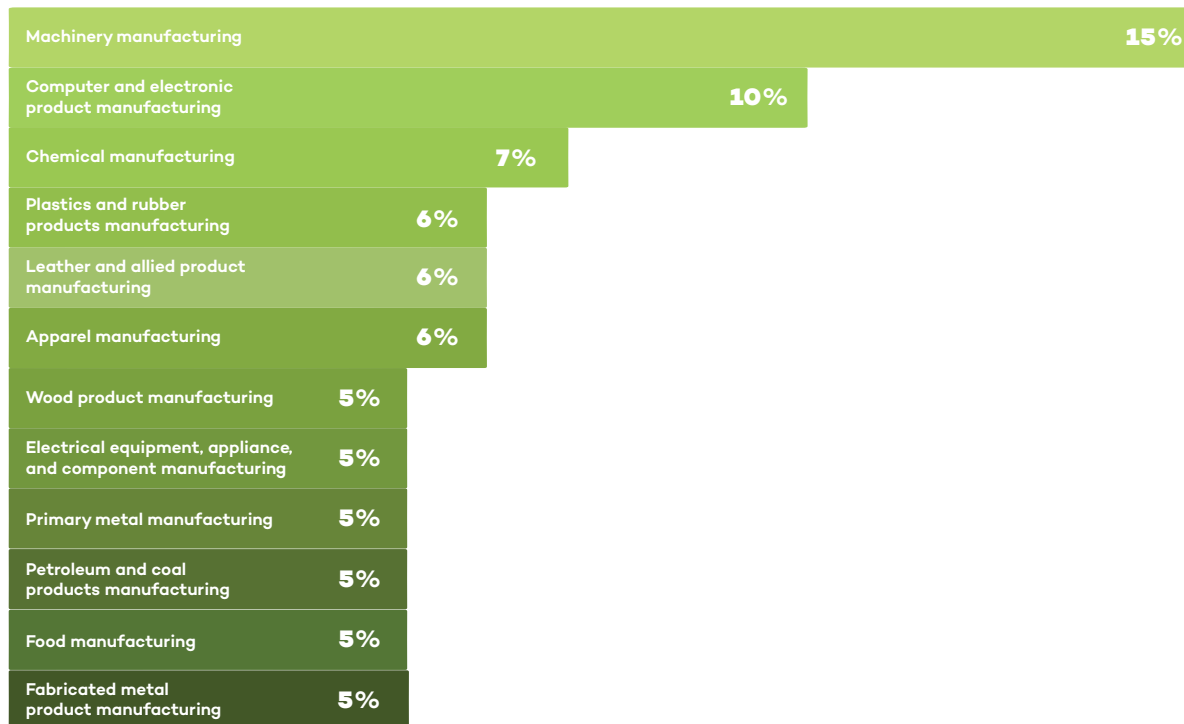
As with any technological transformation, the role of human acceptance is critical. Every company we spoke with stressed that AI represents a means to make their employees more productive and offer them more fulfilling tasks. It is critical for leaders to recognize the opportunities that AI brings and communicate these opportunities to employees. “Part of the leader’s role is to drive a confidence level into the team and say, ‘this can happen, it will happen, and there will be a culture change,’” Martin Smith of Danfoss Power Solutions stressed.

As Joanna Cooper of Daimler Truck North America put it, “it’s all about upskilling people to add value in a much different way. It’s a fallacy to say that AI won’t eliminate some jobs. It will. But it won’t necessarily eliminate the person’s ability to add value to the organization, and those are two very different things.” It is incumbent upon every executive to stress this point to employees. “It’s important for leaders to communicate the role of AI so that people understand it will impact their world, make their world better, and shift what they will be able to do,” Cooper added.

About the Research

Manufacturers Alliance surveyed more than 200 leaders in manufacturing to better understand the state of AI in manufacturing currently and in the near future. We have highlighted some statistics about the respondents.

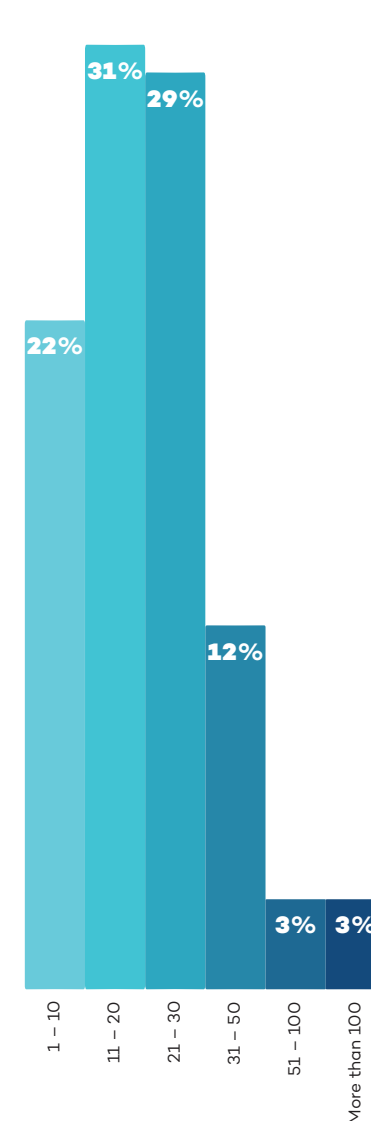
Top Primary Industries



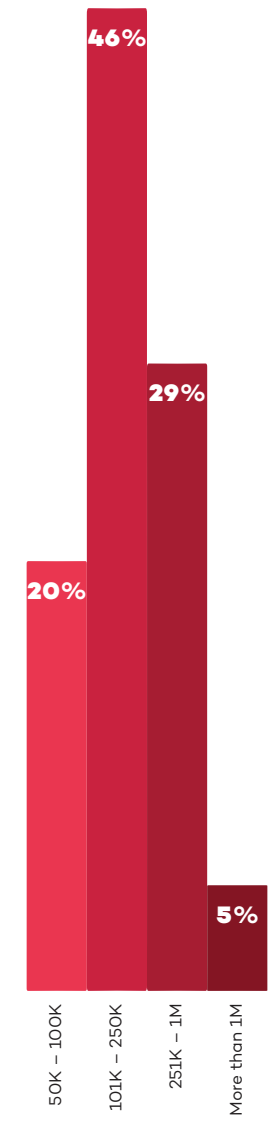
Annual Company Revenue



Number of Factories



Average Factory Size (in Square Footage)



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